



SCUOLA
ALTI STUDI
LUCCA



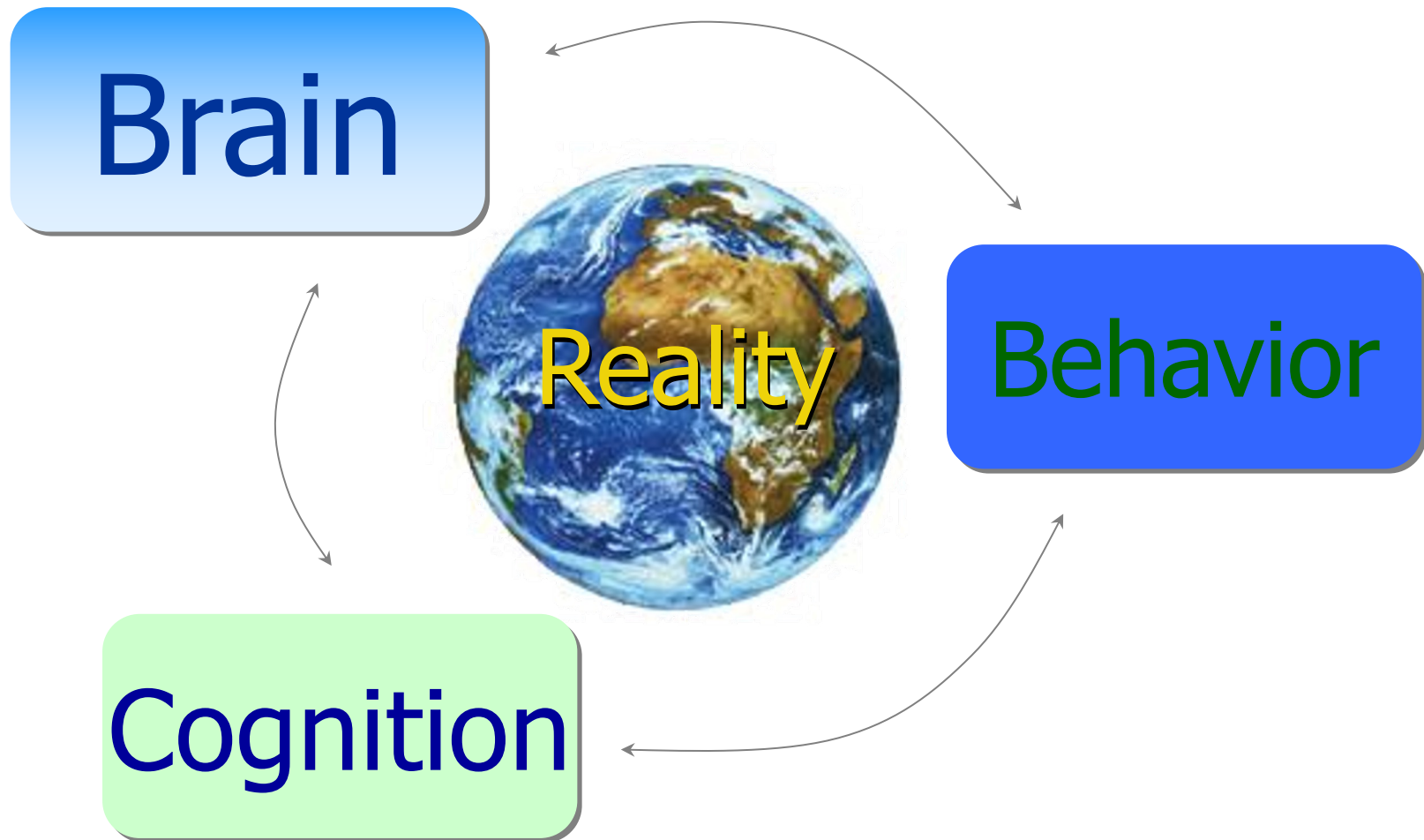
EMOTION AND BEHAVIOR: GENES, BRAIN AND ENVIRONMENT

Pietro Pietrini

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The theoretical framework



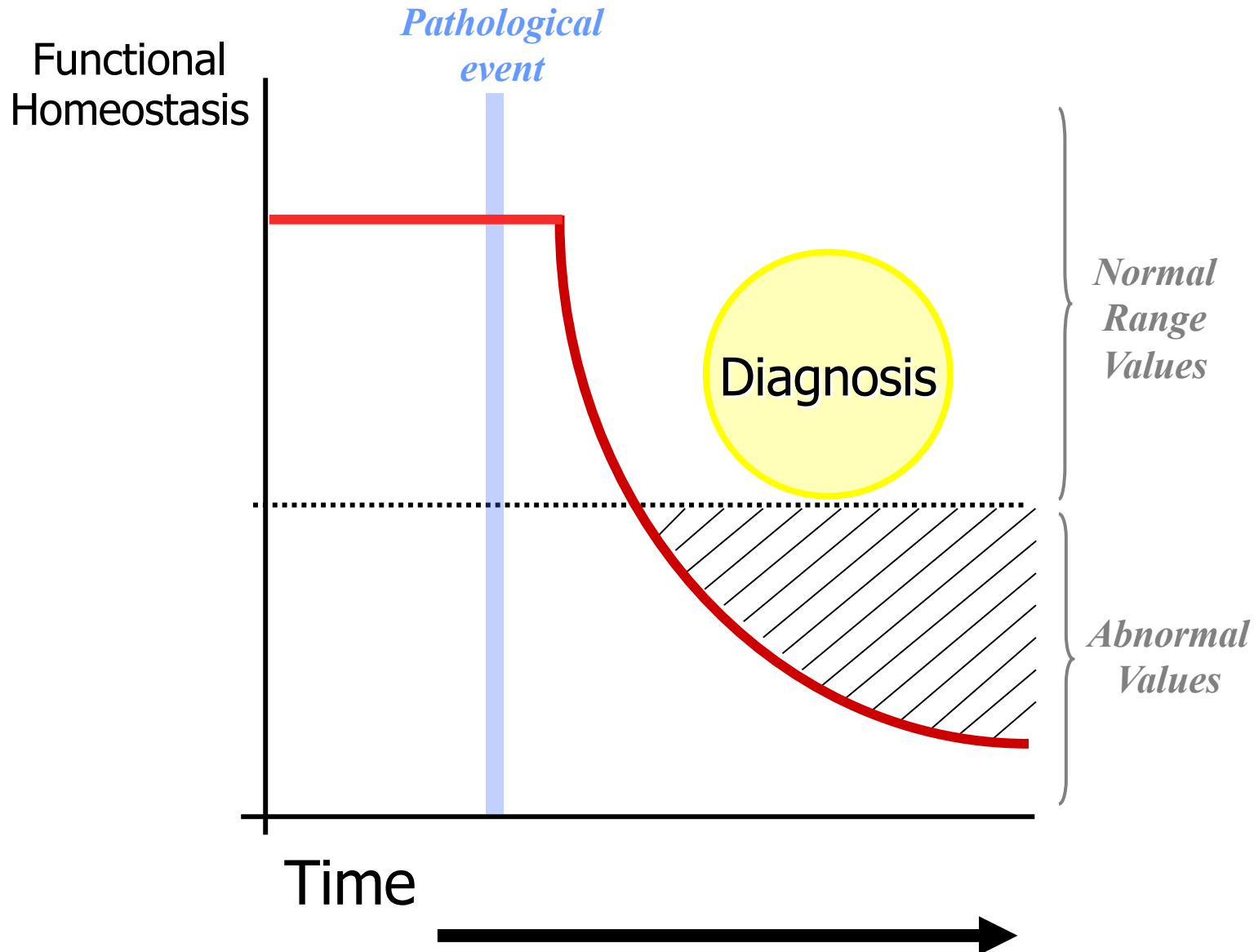
The Team @MOlecular MInd Lab



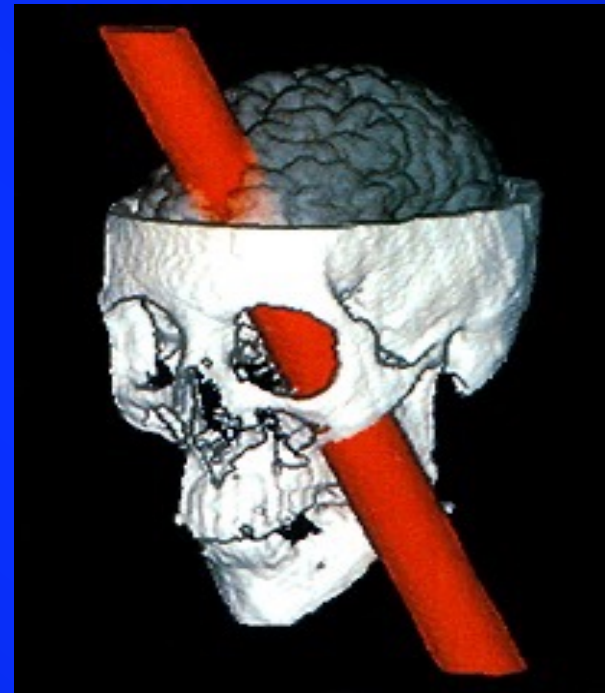
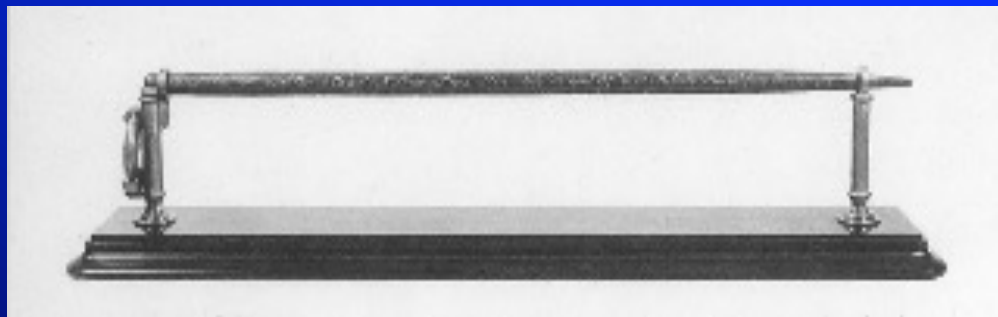
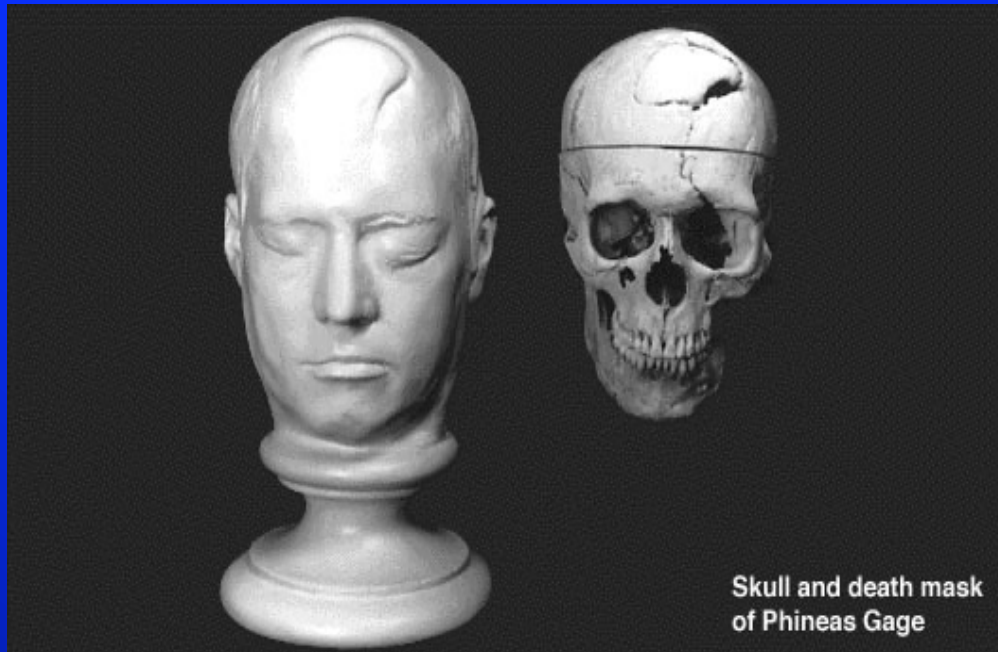
- Emiliano Ricciardi
- Giulio Bernardi
- Gustavo Cevolani
- Luca Cecchetti
- Davide Bottari
- Giacomo Handjaras
- Marta Bortoletto
- Francesca Setti
- Giulia Avvenuti
- Giada Lettieri
- Monica Betta
- Folco Panizza
- Erica Ordali
- All CCSN PhD students @IMT School
- AXES @IMT School
- Collaborations with UNIFI; UNIPD; SAPIENZA; IUSS



Diagnostic Process

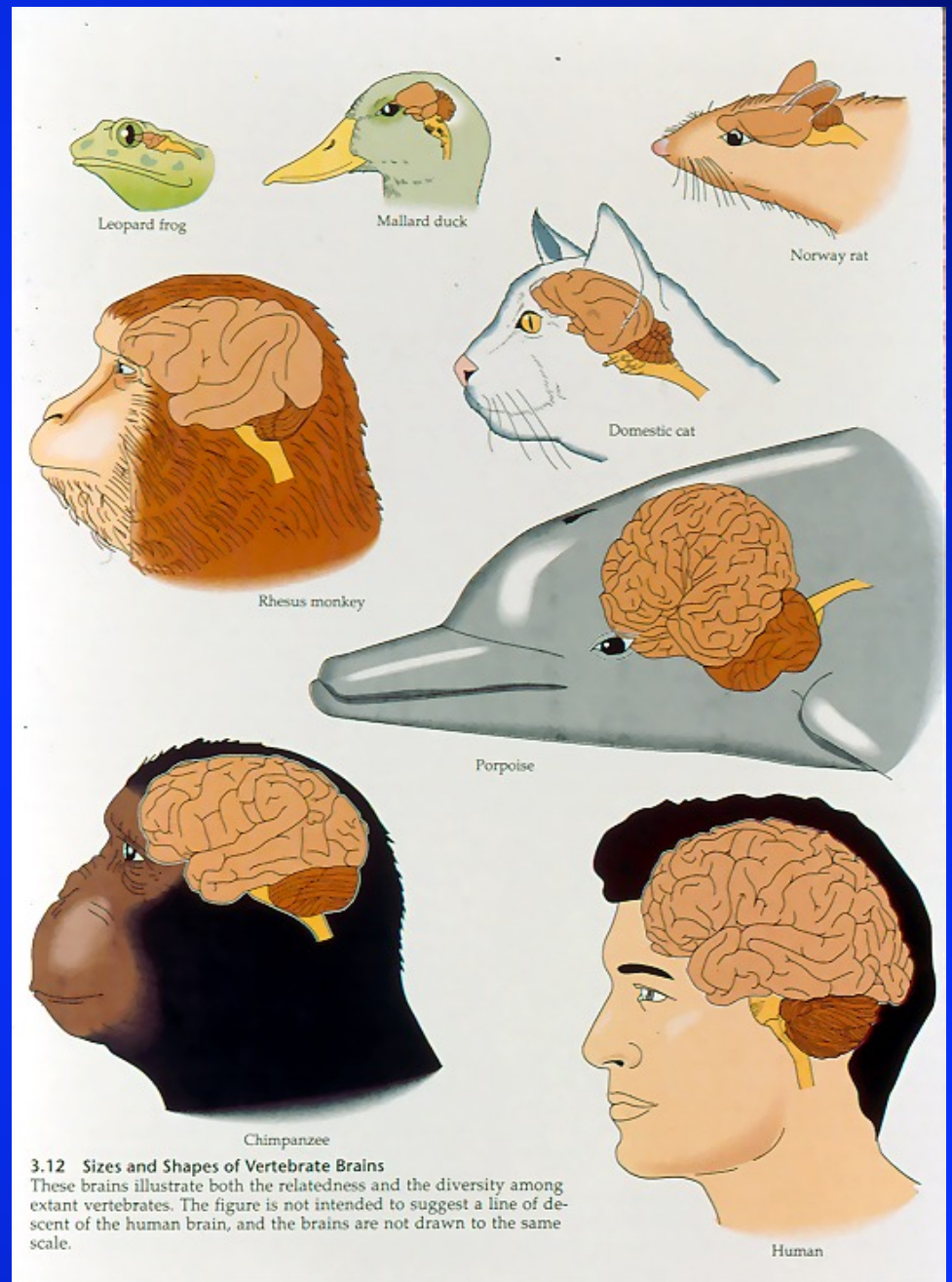


Prefrontal Cortex Lesions and Behavior: the Phineas Gage Case



Searching for the Mind by Looking into the Brain

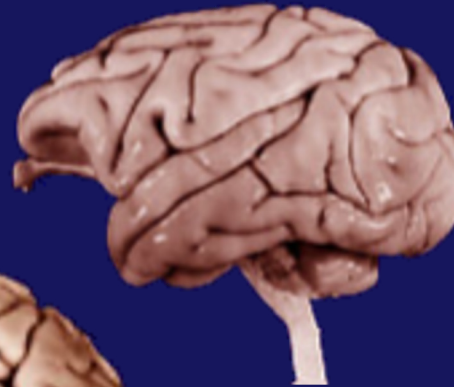
- **Functional Exploration of the Brain:**
a "true window" to observe the brain in action
- **Molecular Biology and Genetics:**
how our genes (DNA) shape our destiny



Owl Monkey



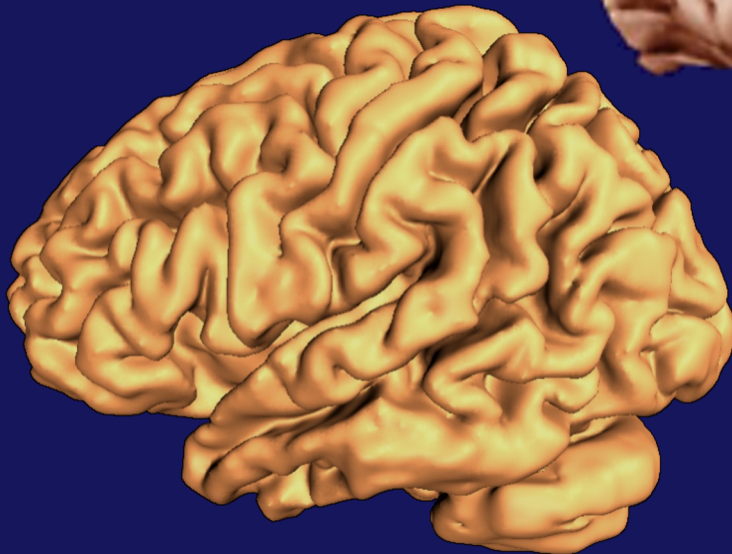
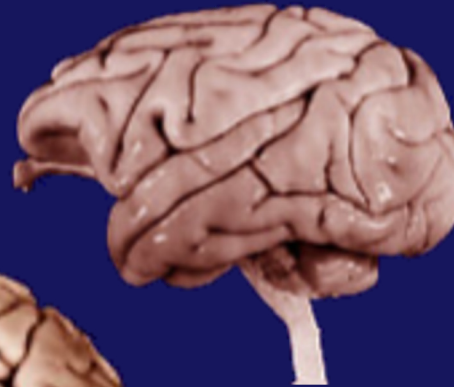
Mandrill baboon



Mandrill baboon



Owl Monkey



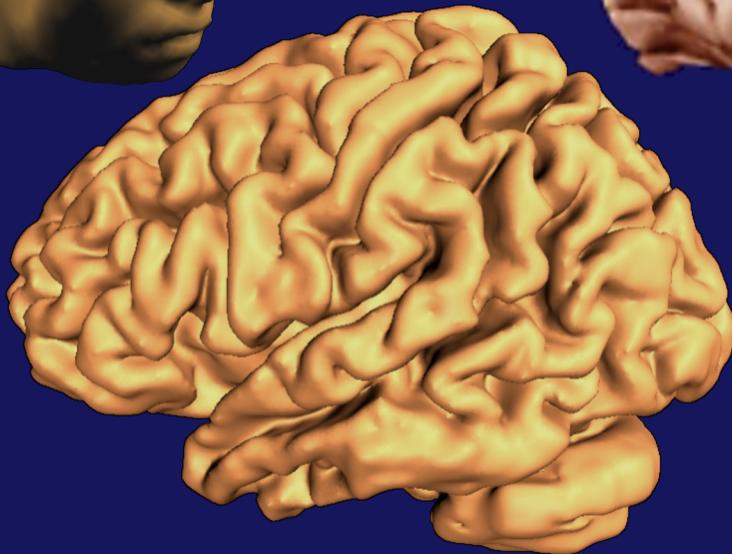
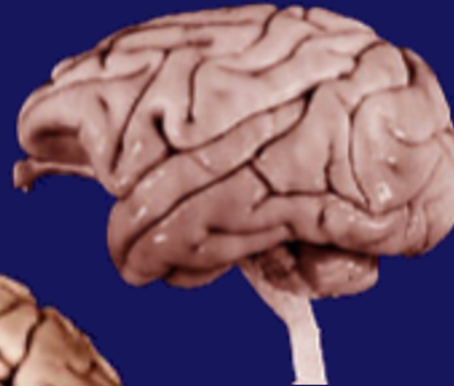
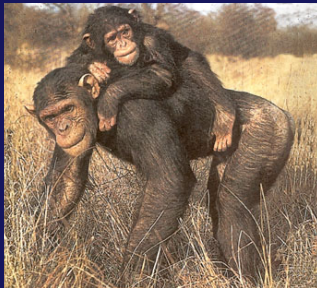
Owl Monkey



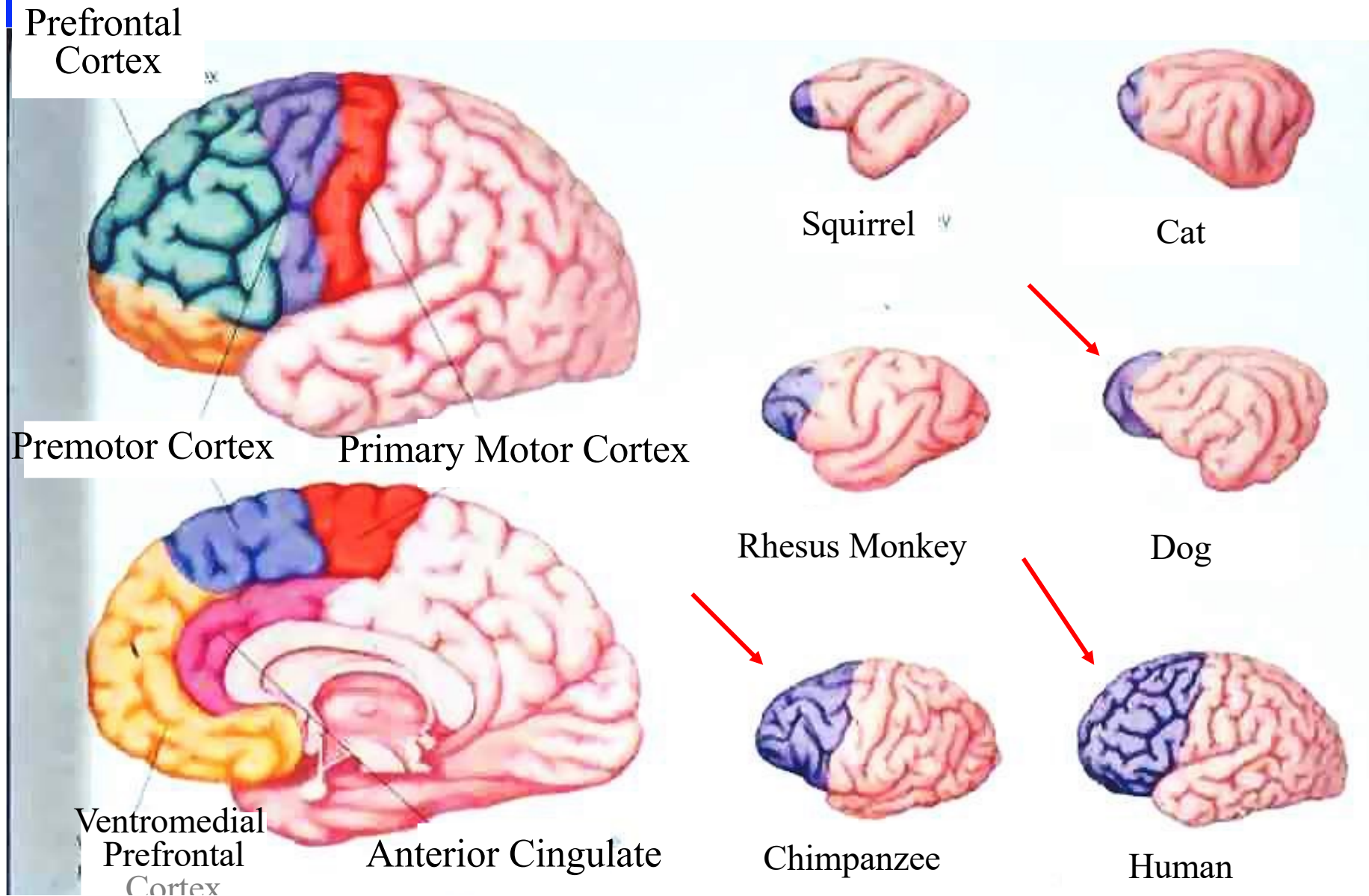
Mandrill baboon



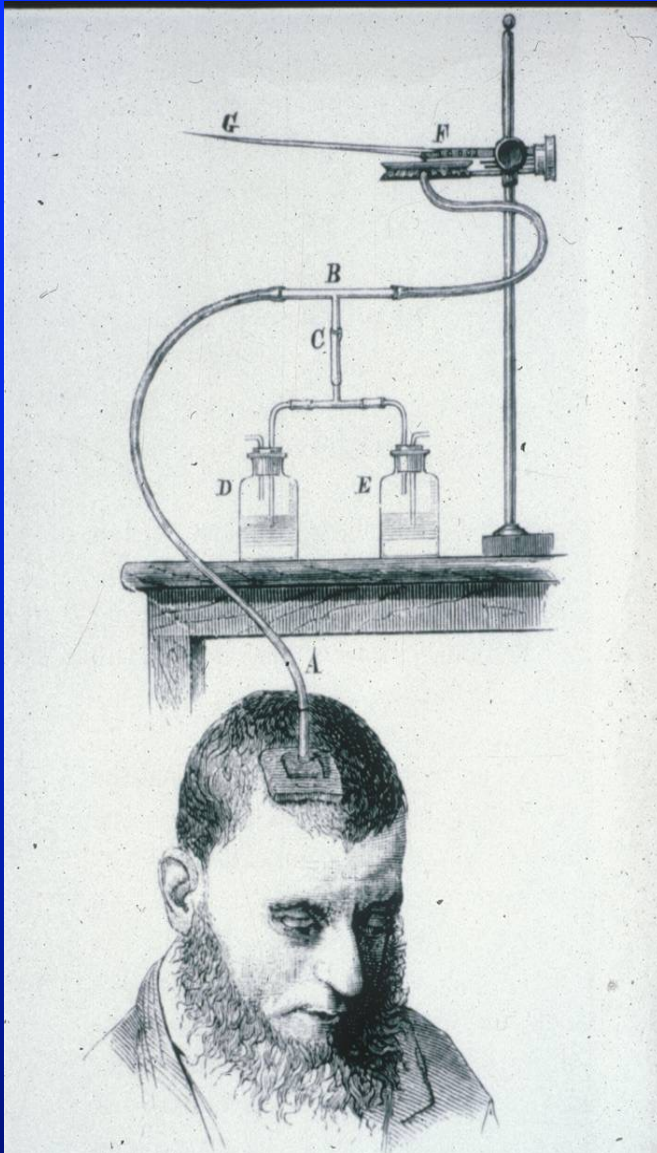
Chimpanzee



Brain Development and Functional Organization



The Mosso's experiment: Changes in Mental Activity are associated with Increases in Regional Cerebral Blood Flow



A: Arm C: Cerebrum

Resting quiet

1 { A
C

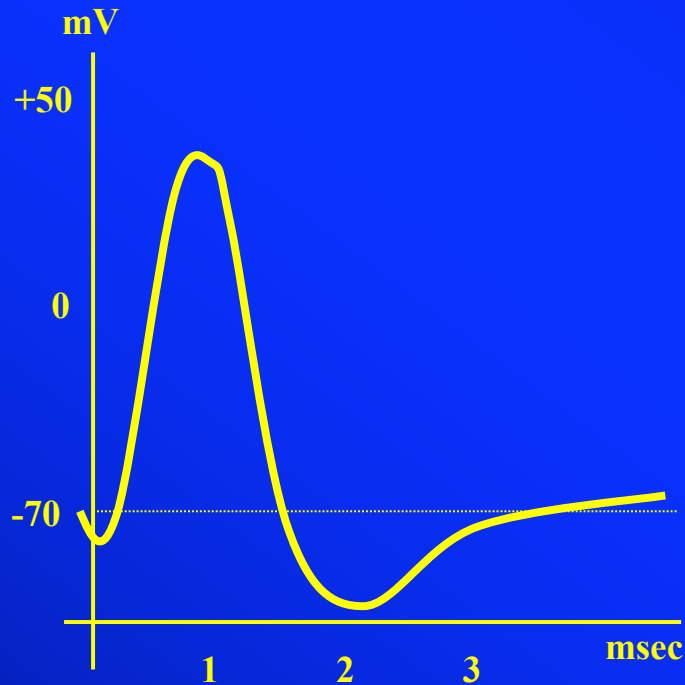
Arrow: The Church Bells ring Noon

2 { A
C

Arrow: should we recite the Ave Maria?

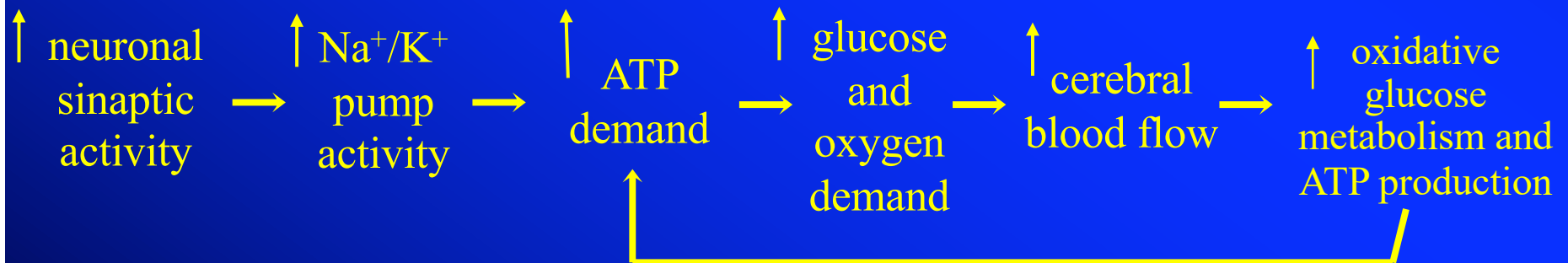
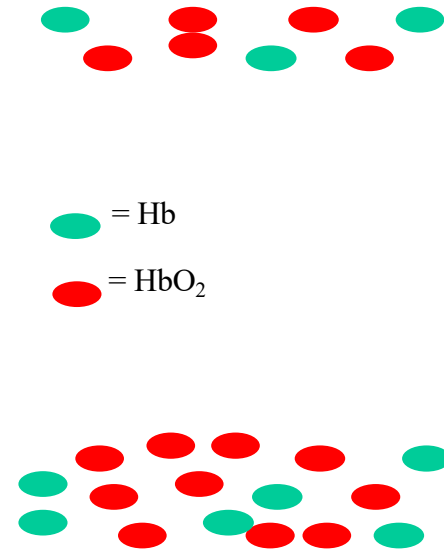
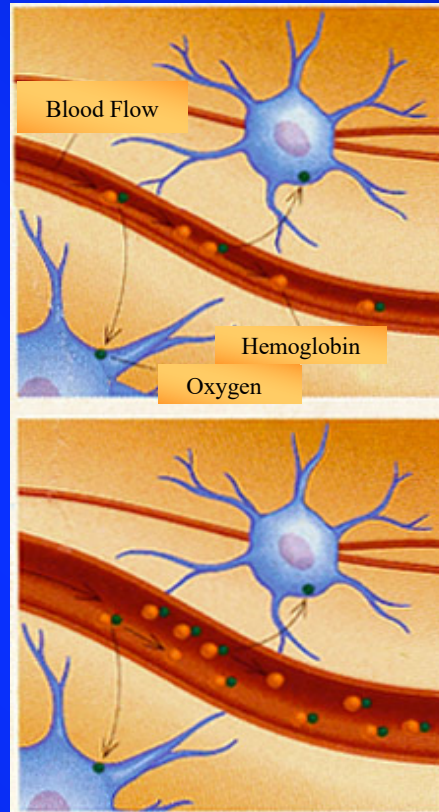
3 { A
C

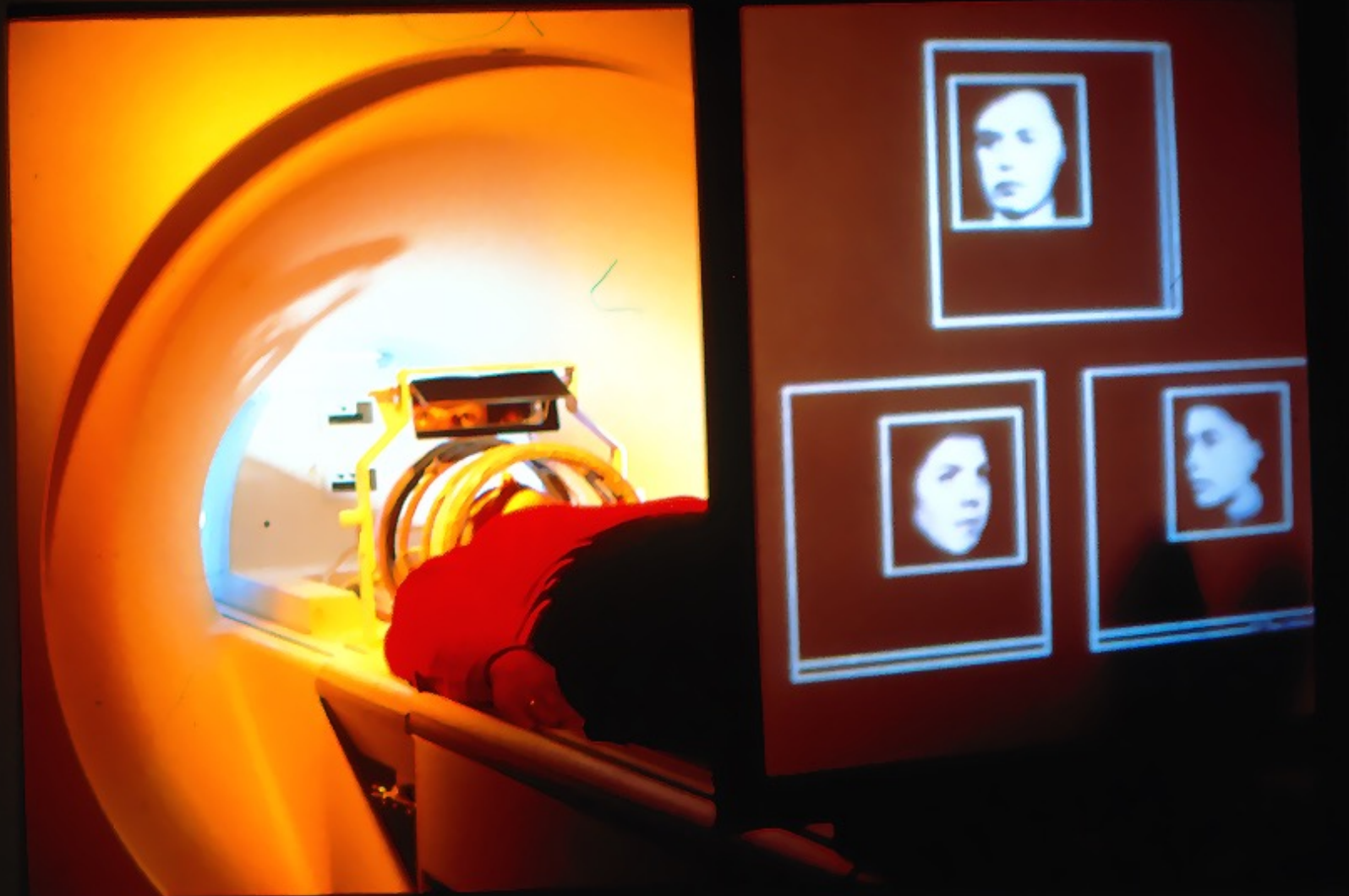
In vivo functional exploration of the metabolic and blood flow correlates of neuronal activity



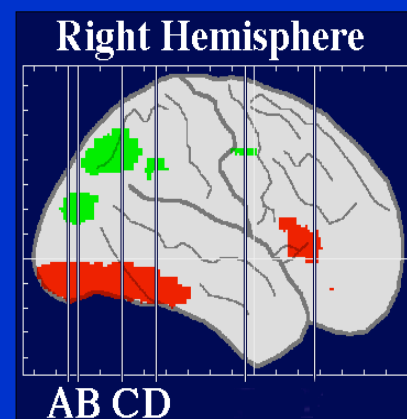
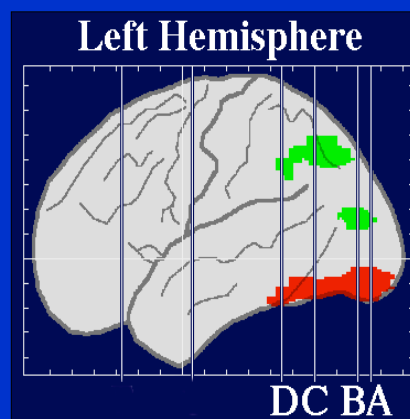
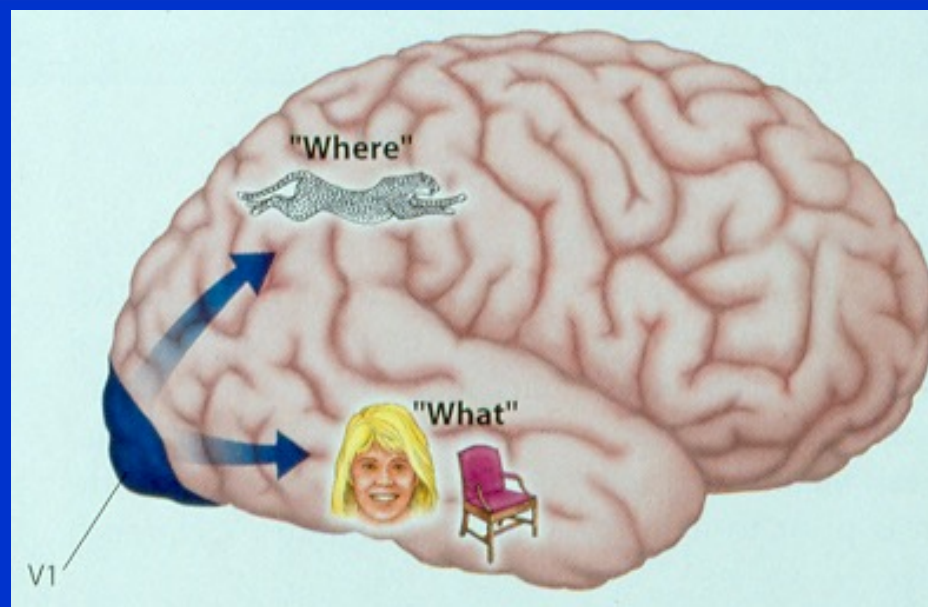
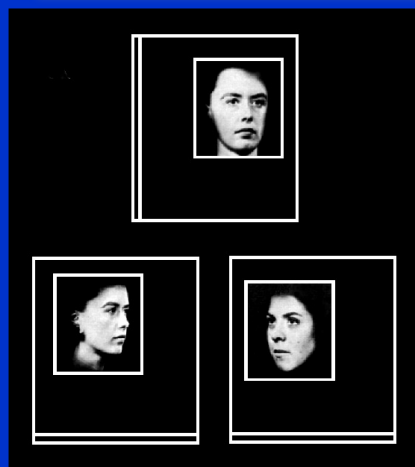
Rest

Activation

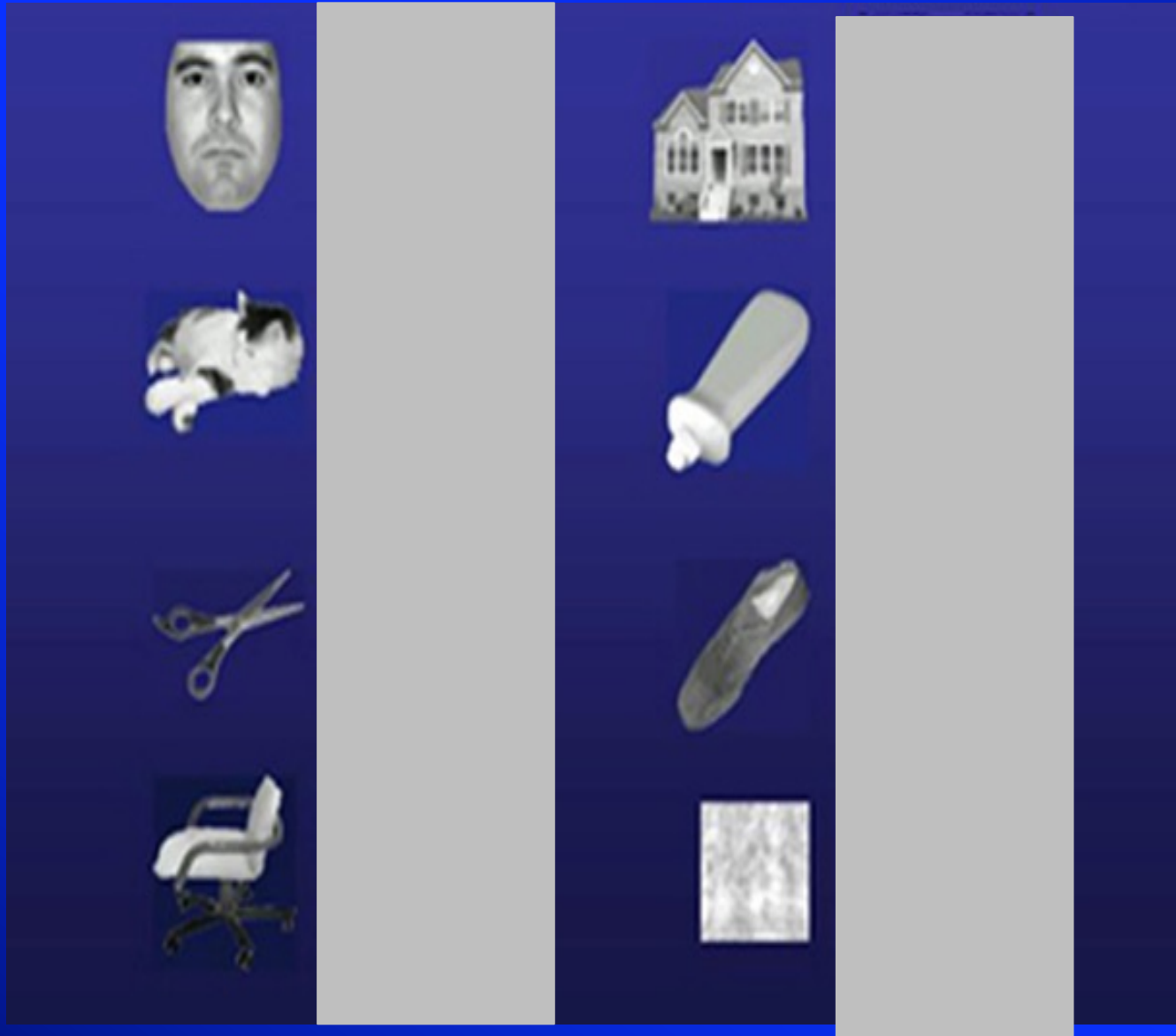




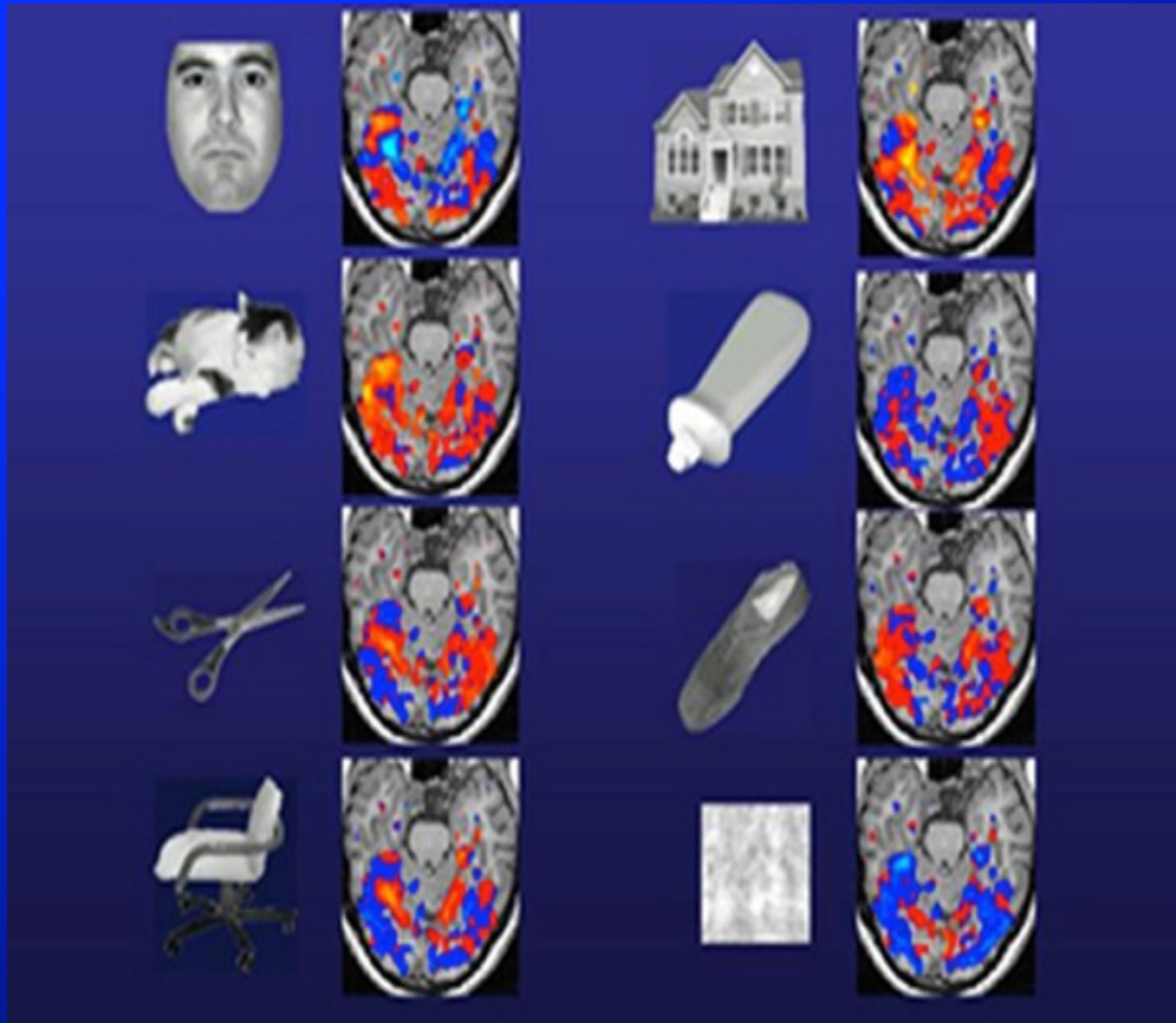
Perception of the External World in the Brain



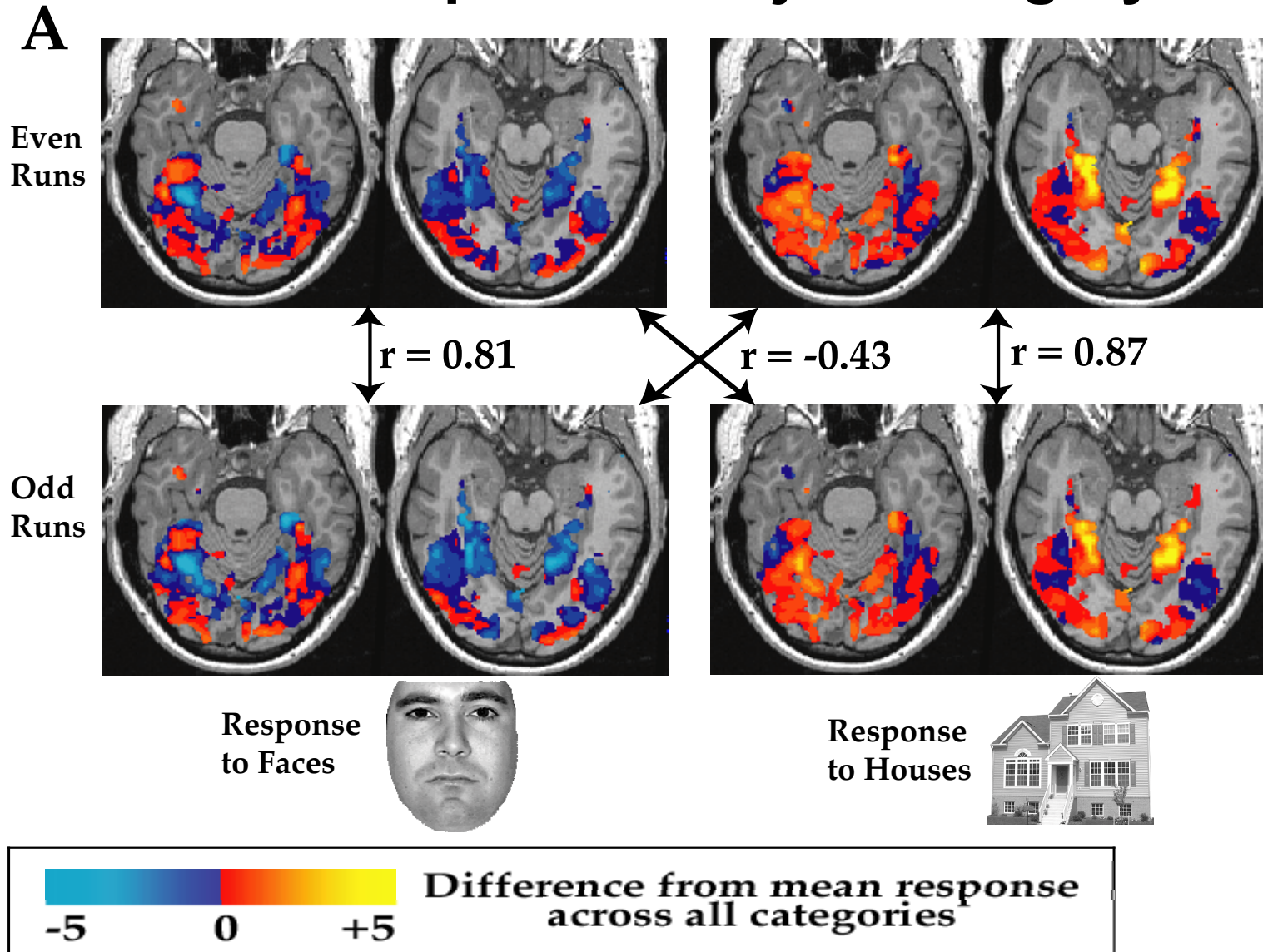
How does the Brain recognize distinct object categories?



Distributed and overlapping patterns of neural response across the ventro-temporal cortex



Pattern of neural response is highly specific and predicts object category

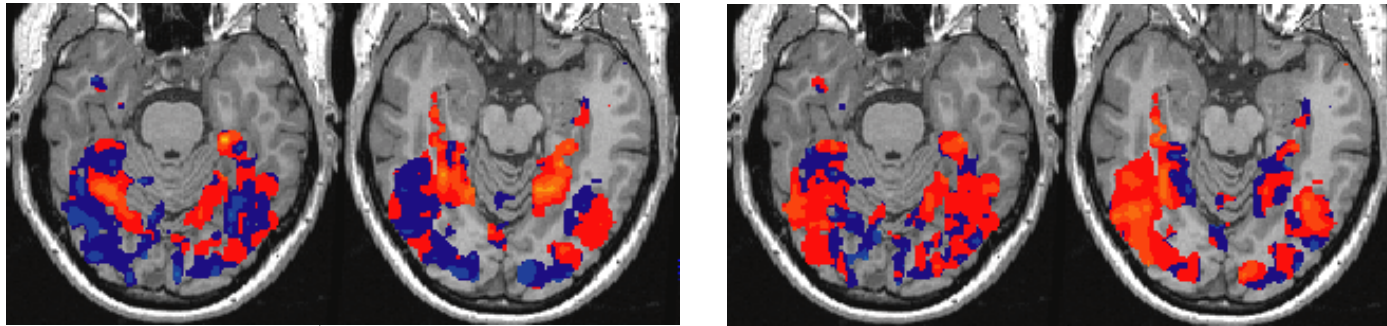


Haxby et al., *Science*, 2001

Pattern of neural response is highly specific and predicts object category

B

Even
Runs

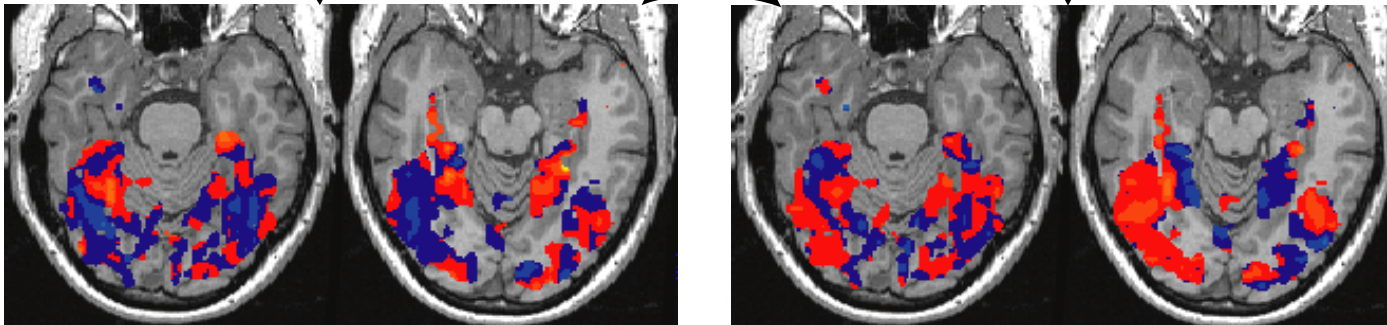


$r = 0.45$

$r = -0.11$

$r = 0.55$

Odd
Runs



Response
to Chairs



Response
to Shoes



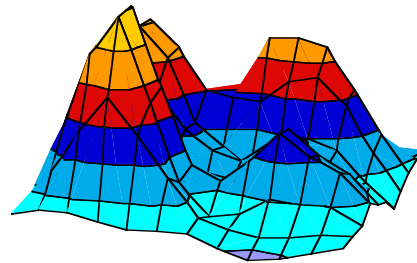
Difference from mean response
across all categories

Haxby et al., *Science*, 2001

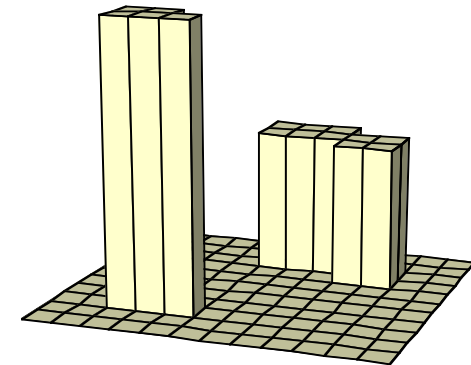
Object Form Topology in the Ventro-Temporal Cortex



Distributed Representation 'Object Form Topology'

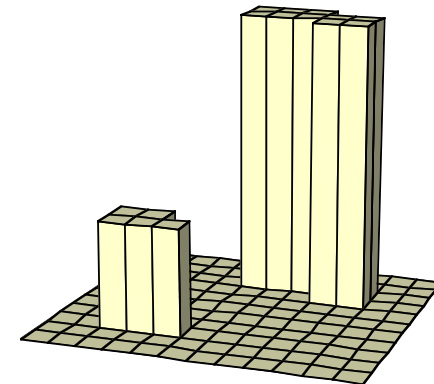
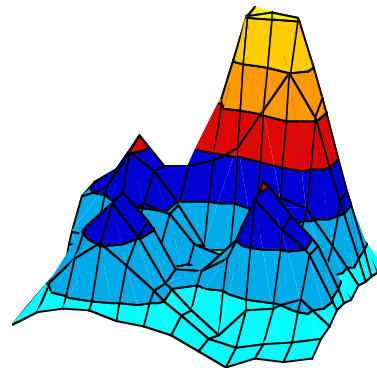


Category-Selective Regions



Distributed and Overlapping Representations of Faces and Objects in Ventral Temporal Cortex

James V. Haxby,^{1*} M. Ida Gobbini,^{1,2} Maura L. Furey,^{1,2}
Alumit Ishai,¹ Jennifer L. Schouten,¹ Pietro Pietrini³

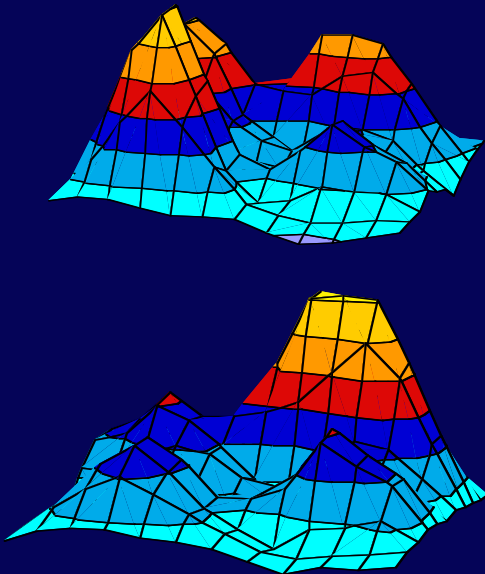


2001

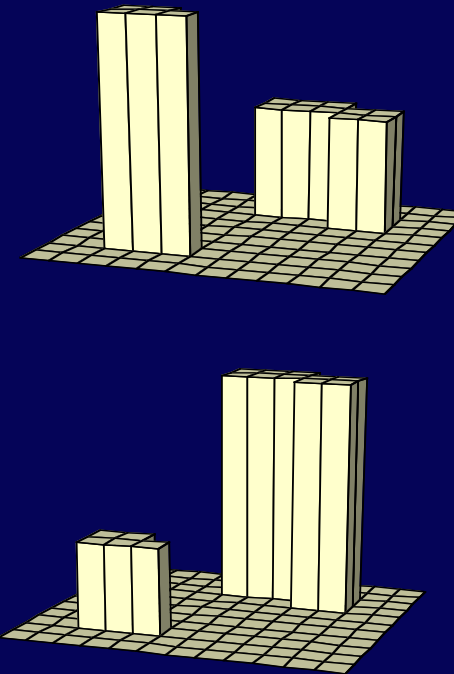
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Distributed Representation 'Object Form Topology'

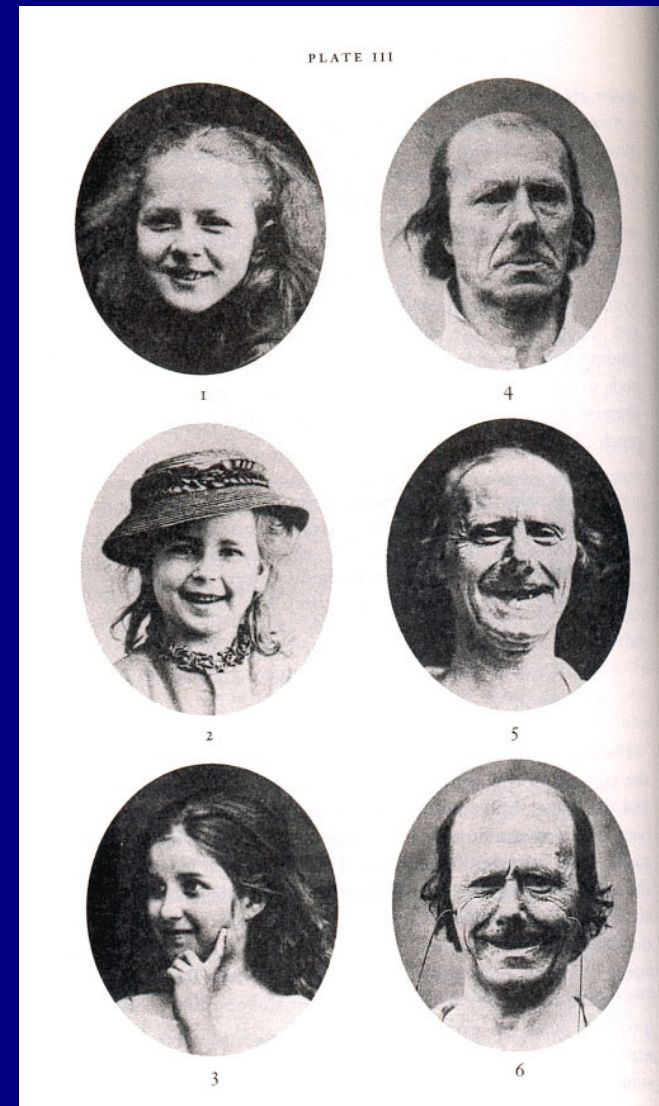
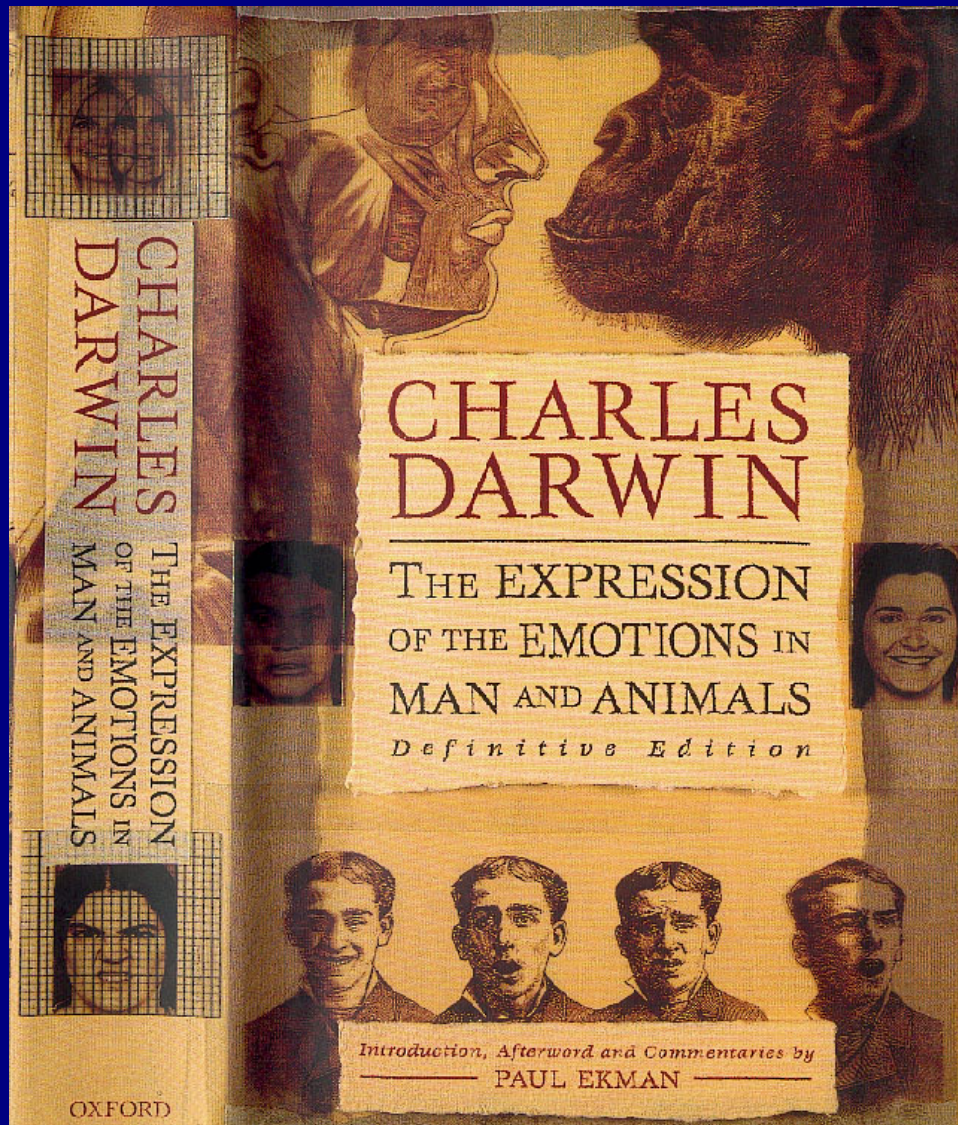


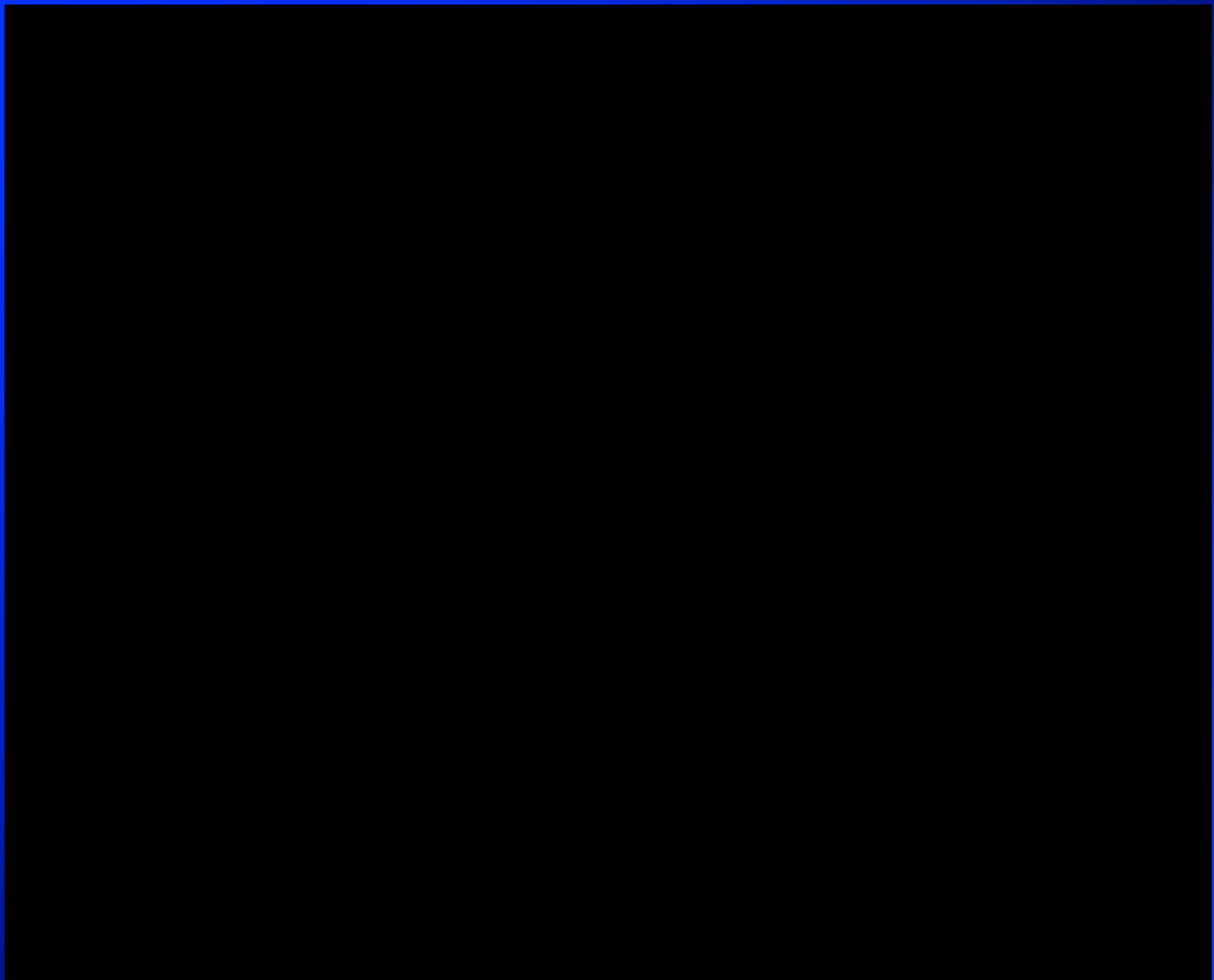
Category-Selective Regions



2001

Face expressions deliver rapid, non-verbal messages-->
crucial for survival



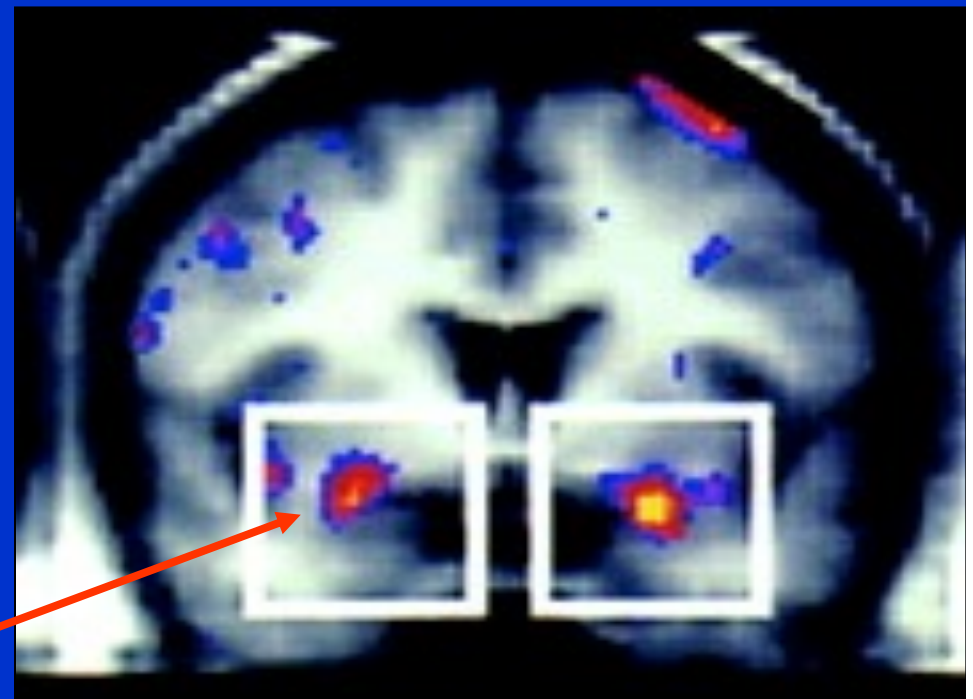


Fearful face perception activates Amygdala

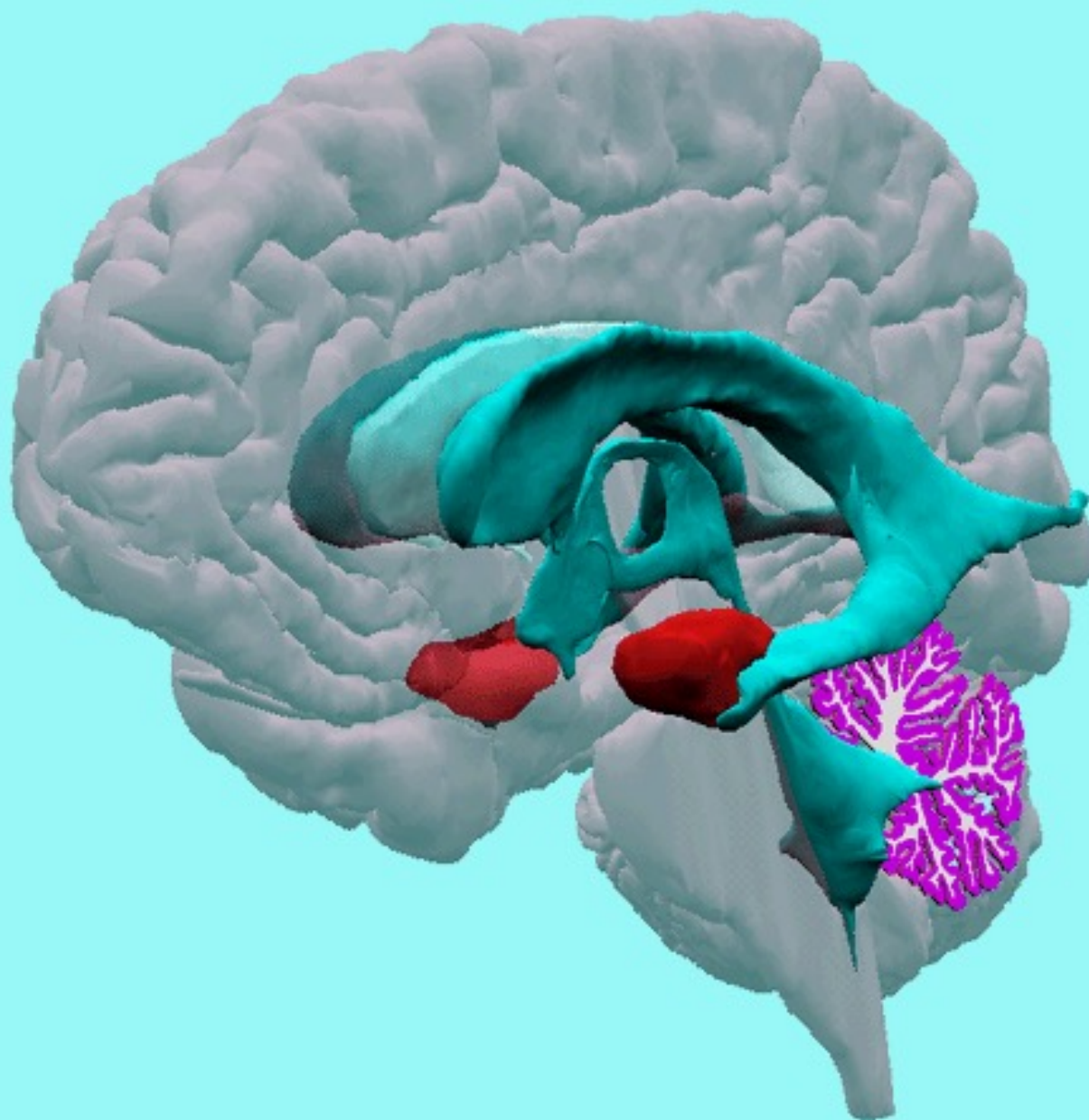
- Visual presentation of neutral or fearful face expressions



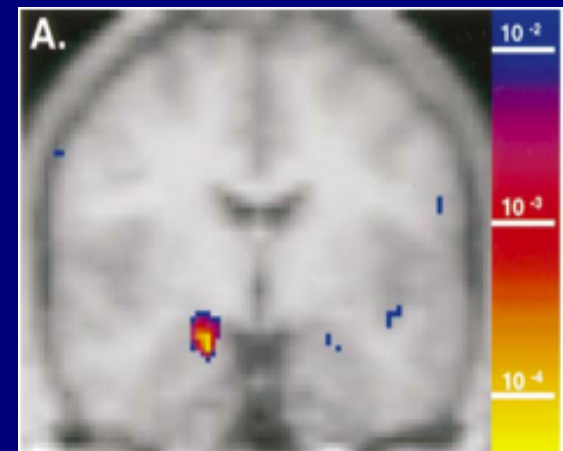
VS



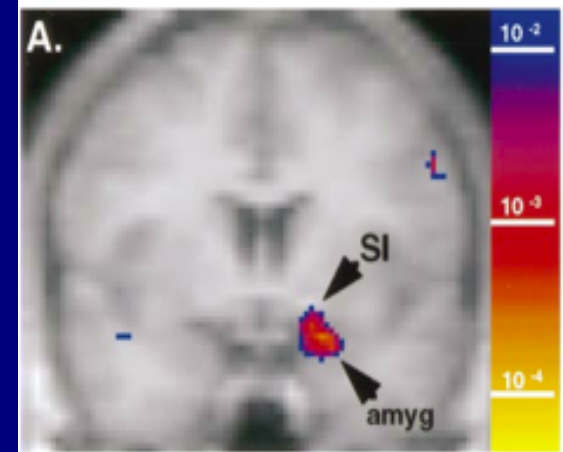
Amygdala



The Fear Instinct



y = -6

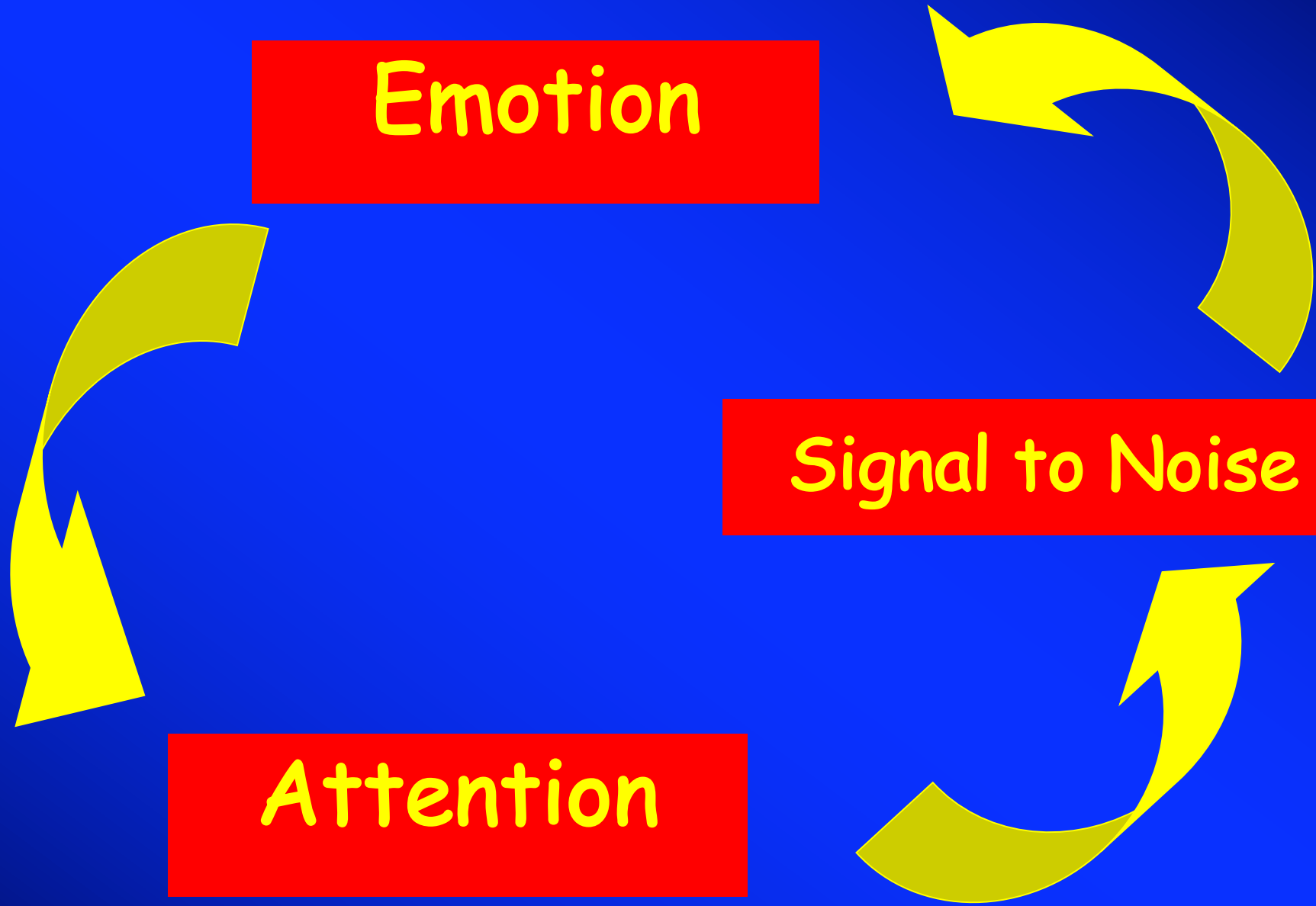


y = 0

Emotion

Signal to Noise

Attention

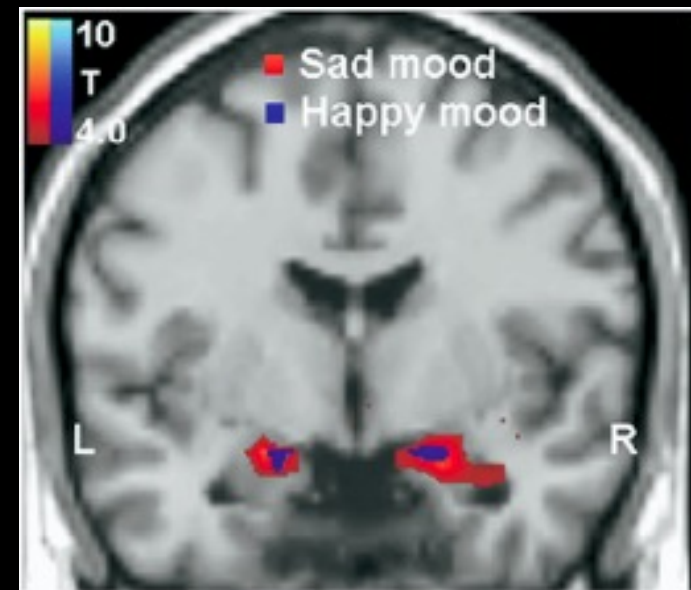
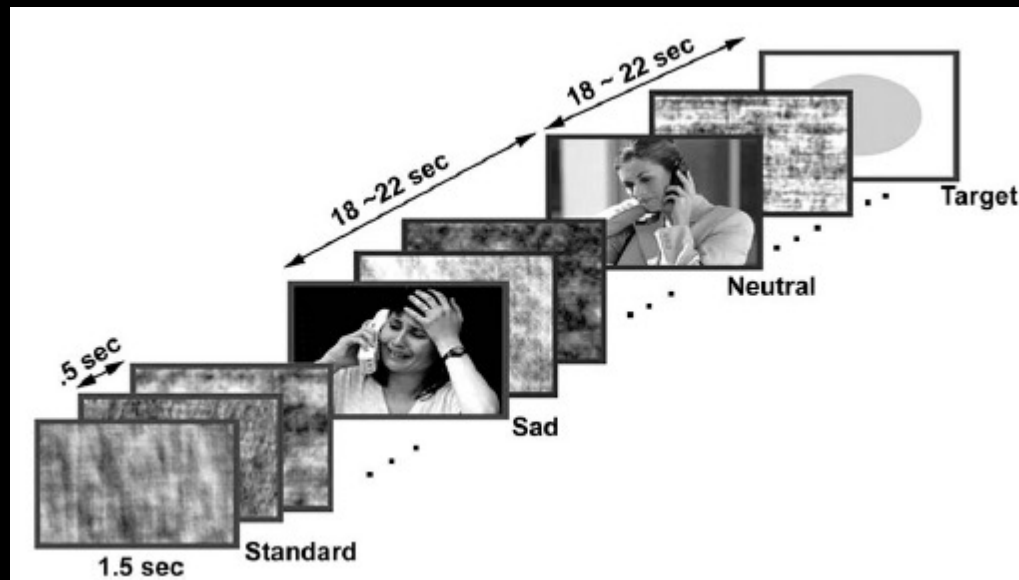




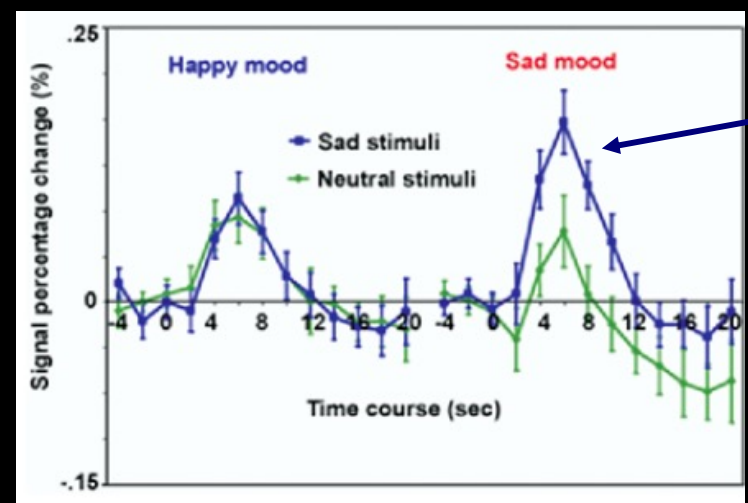
How Mood modulates Brain Response to Emotional Stimuli

Mood Alters Amygdala Activation to Sad Distractors During an Attentional Task

Lihong Wang, Kevin S. LaBar, and Gregory McCarthy



BIOL PSYCHIATRY 2006;xx:xxx
© 2006 Society of Biological Psychiatry



vol. no.

15/2

march-april

1997

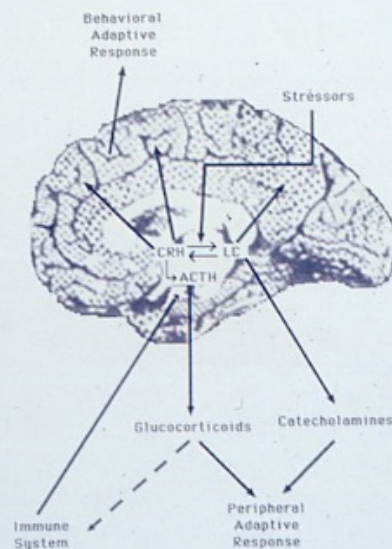
Clinical and Experimental
RHEUMATOLOGY

Editorial

Life events in the course of chronic diseases:
A psychological myth or a psycho-neuro-biochemical loop ?

P. PIETRINI, MD, PH.D.^{1,2}

M. GUZZELLI, MD³

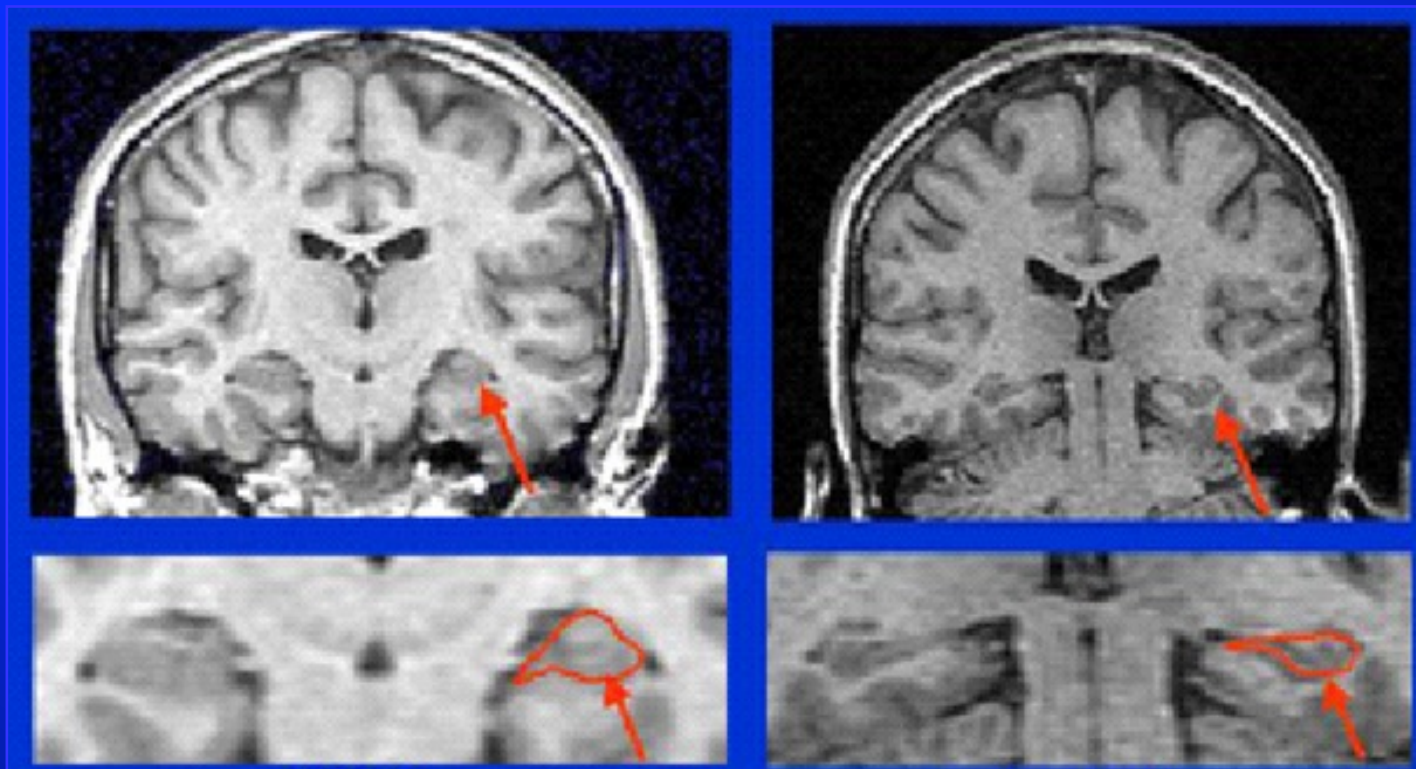


Die Lebensmüden - Ferdinand Hodler
Neue Pinakothek, München



Brain Atrophy in Depression?

Atrophy of the Hippocampus in Depression

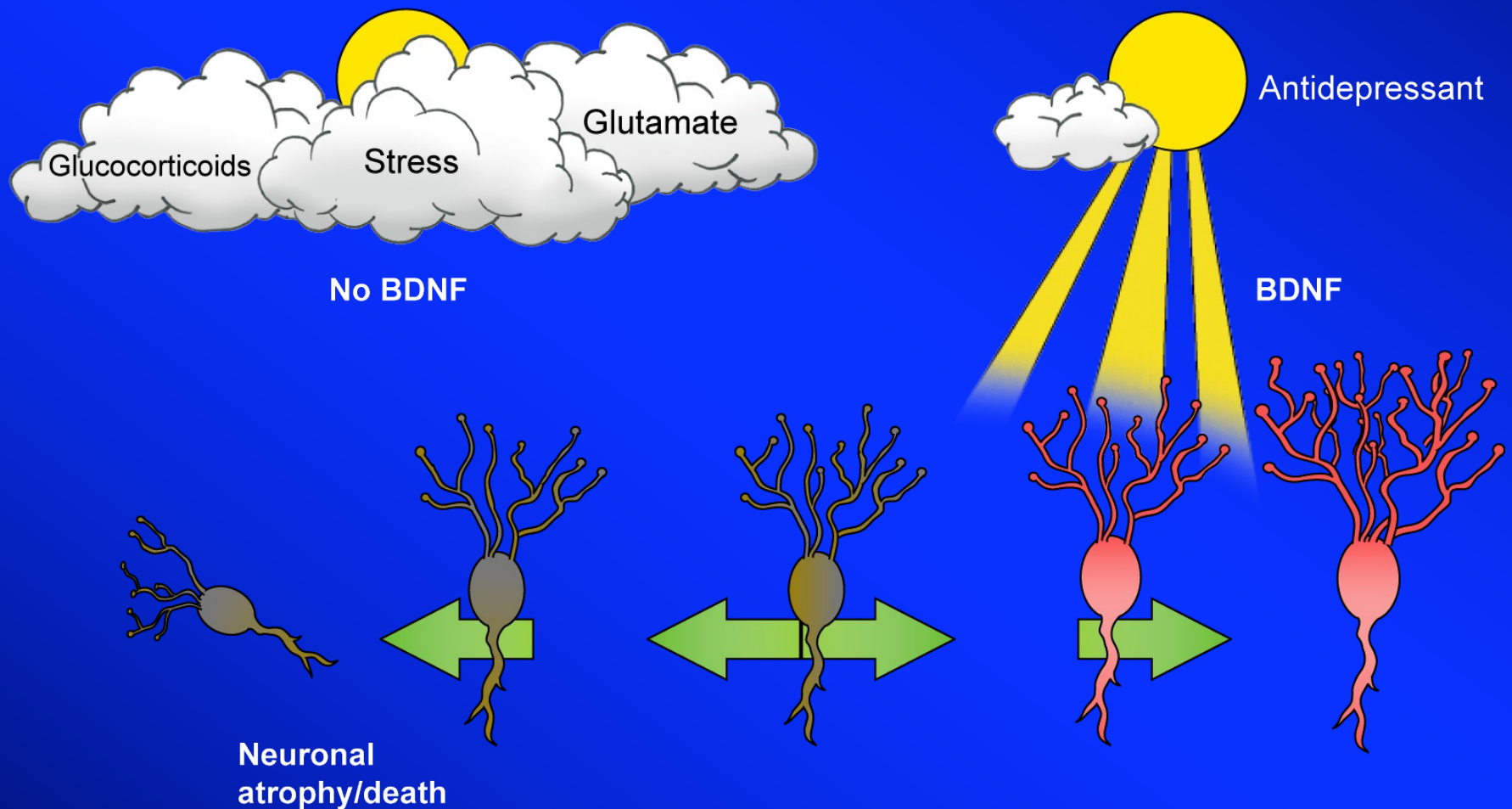


Normal

Depression

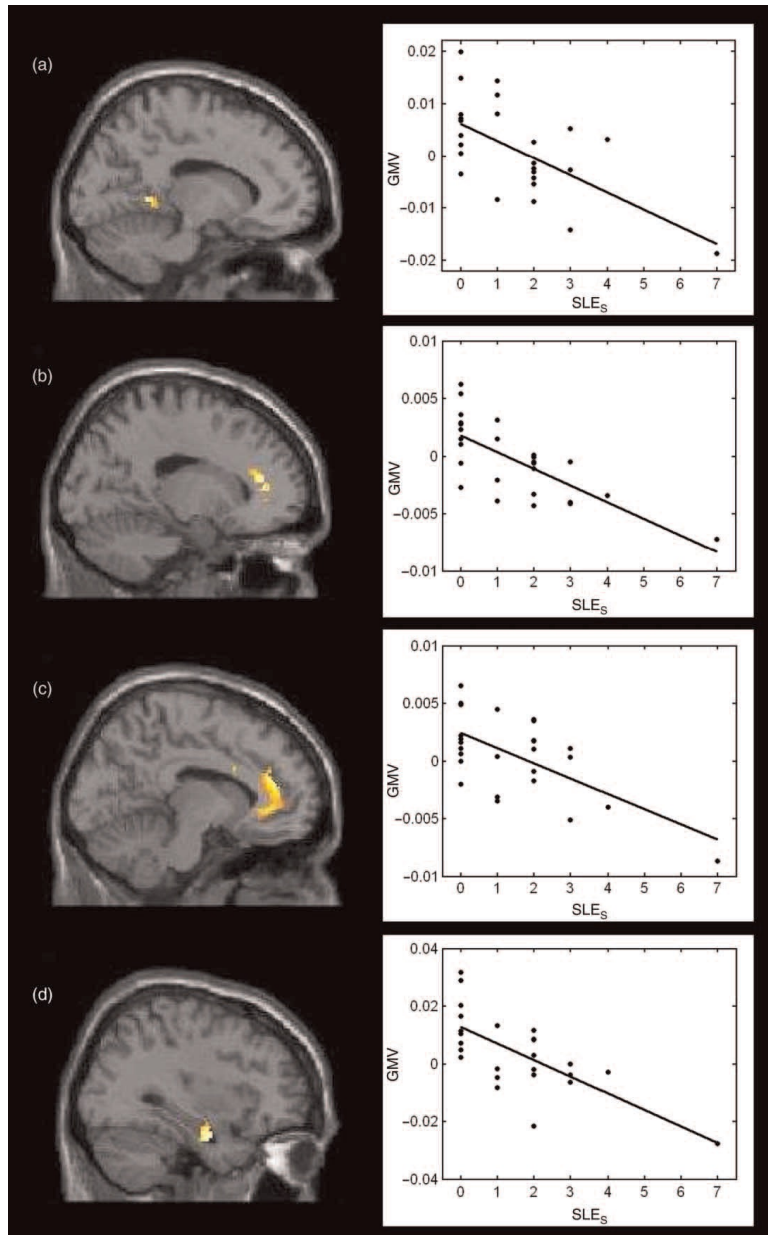
Bremner JD, et al. *Am J Psychiatry* 2000;157(1):115-118.
Reprinted with permission from JD Bremner.

The Monoamine Hypothesis of Gene Action: The Impact of Stress on BDNF



Stahl SM. *Essential Psychopharmacology: Neuroscientific Basis and Practical Applications*; 2000:187.

Life Events Affect Hippocampal Volume



Number of life events over a year period is inversely correlated with hippocampal volume

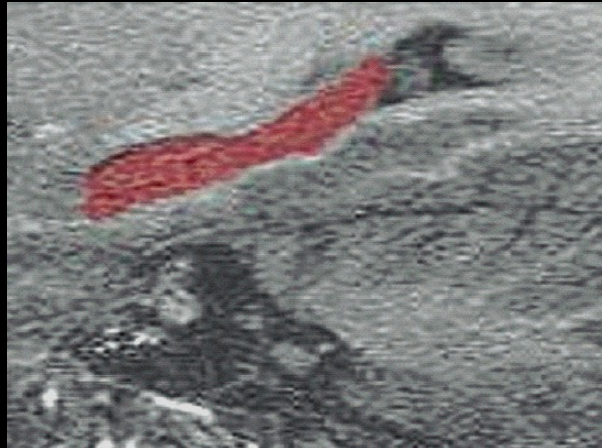
Papagni et al., 2011

DEBATES IN NEUROSCIENCE

Depression: A Case of Neuronal Life and Death?

Ronald S. Duman

Healthy control subject



Patient with recurrent depression



Campbell & MacQueen, 2002

Neurogenesis in Adult Brain

Gould et al 1999; Gage 2000; Duman et al 2001

Stress Decreases Adult Neurogenesis (↑HPA axis)

Gould et al 1997; Gould et al 1998; Tanapat et al 2000;
Czeh et al 2001, 2002; van der Hart et al 2002; Pham et
al 2003; Malberg and Duman 2003; Alonso et al 2004

Hippocampus (volume ↓) - Depression (↑ HPA axis)

File et al 2000; Menard and Treit 2001; Drevets 2001;
Manji et al 2001; Deacon et al 2002; Degroot and Treit
2002

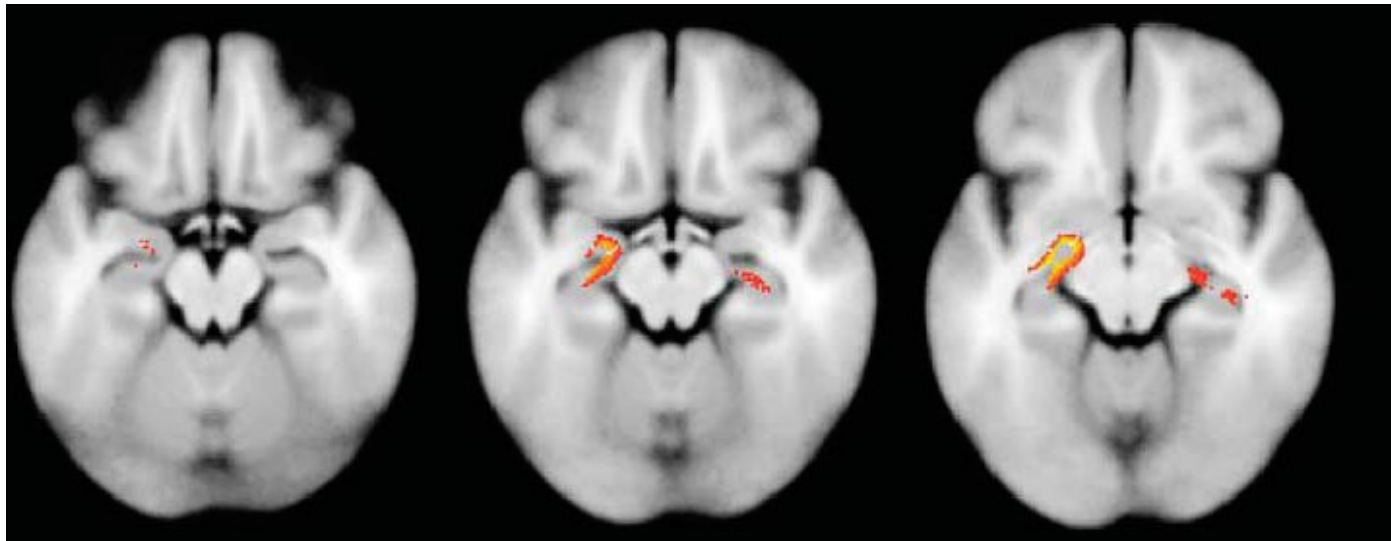
Antidepressant Treatment Increases Adult Neurogenesis

Madsen et al 2000; Malberg et al 2000; Czeh et al 2001;
Manev et al 2001; Santarelli et al 2003

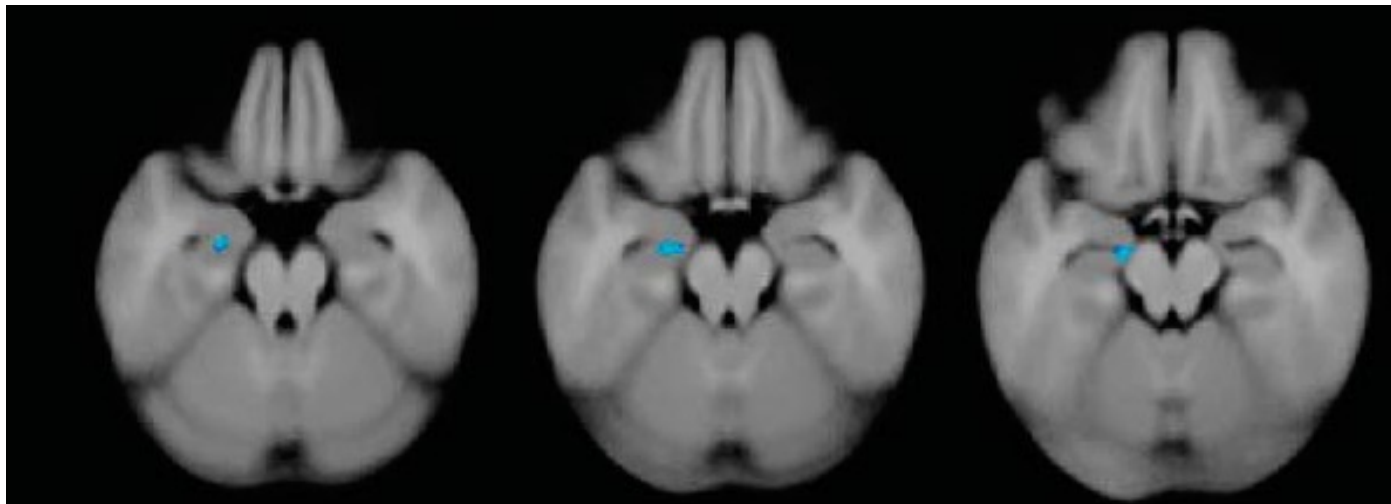
Antidepressant Treatment Blocks the Effects of Stress on Adult Neurogenesis

Lee et al 2000; Czeh et al 2001; van der Hart et al 2002;
Malberg and Duman 2003; Alonso et al 2004

Hippocampal Volume and Depression



pts. < controls



After 3 years

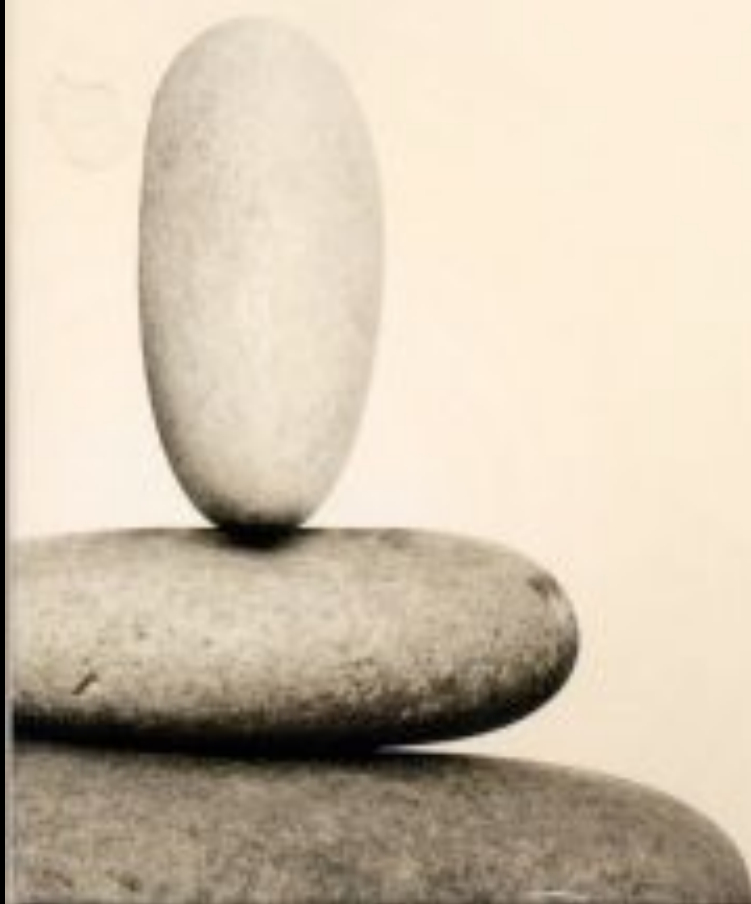
Recovered patients
< non remitted pts

Fredl et al., 2008

Author of Coming to Our Senses & Wherever You Go, There You Are

JON KABAT-ZINN

MINDFULNESS FOR BEGINNERS



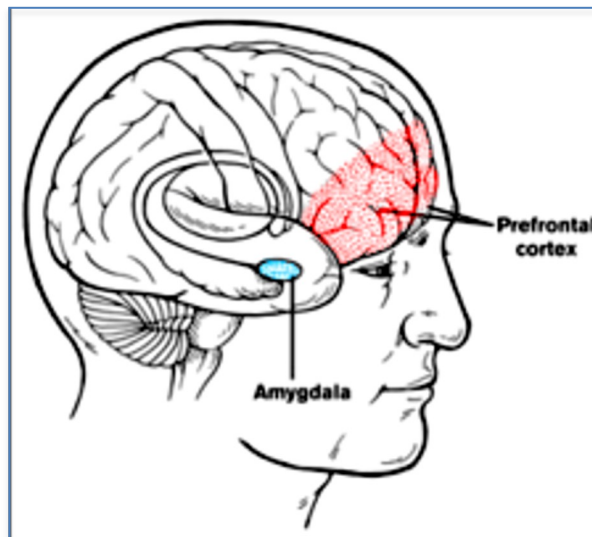
General Effects of Mindfulness Based Stress Reduction (MBSR)

Reduction:

- social anxiety;
- depressive symptoms;
- aggressivity;
- Stress perception
- Cognitive reactivity;
- Stress-related morbidity.

Increase:

- Mood;
- Spirituality;
- Empathy;
- Awareness;
- Life satisfaction;
- Life quality.





“Relaxation Response”



Integrated physiologic response characterized by:

- ↓ **O₂ consumption**
- ↓ **CO₂ removal**
- ↓ **Heart rate and respiratory frequency**
- ↓ **Arterial blood pressure**
- ↓ **Haematic lactate**
- ↑ **Skin resistance**



Shift of the sympatho-vagal balance toward parasympathetic predominance

Mindfulness meditation and EEG

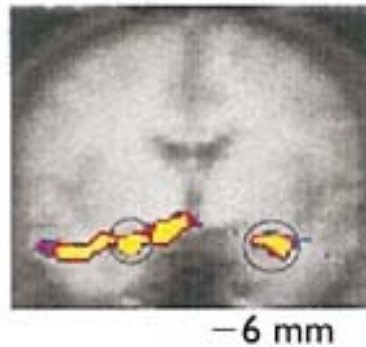


↑ Teta
↑ Alfa

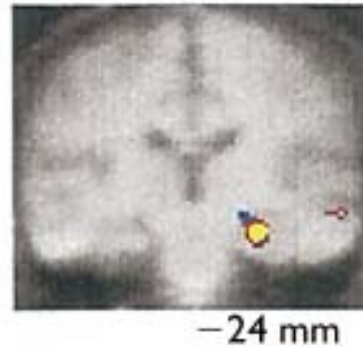
Brain activity and Meditation

regioni più attive nei soggetti in meditazione rispetto a soggetti rilassati

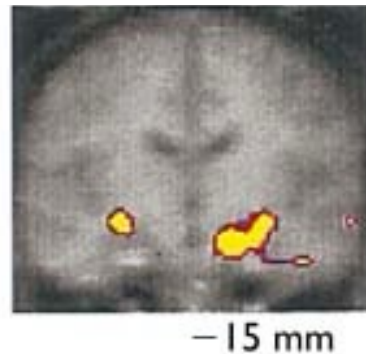
amigdala



ippocampo

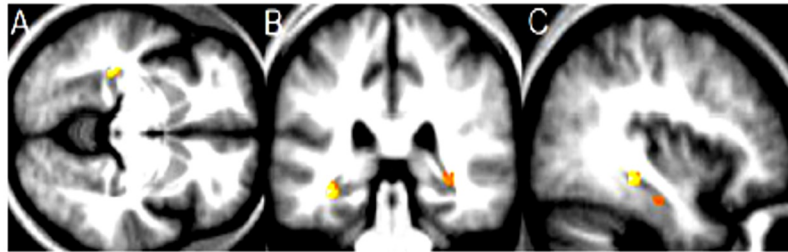


basal ganglia

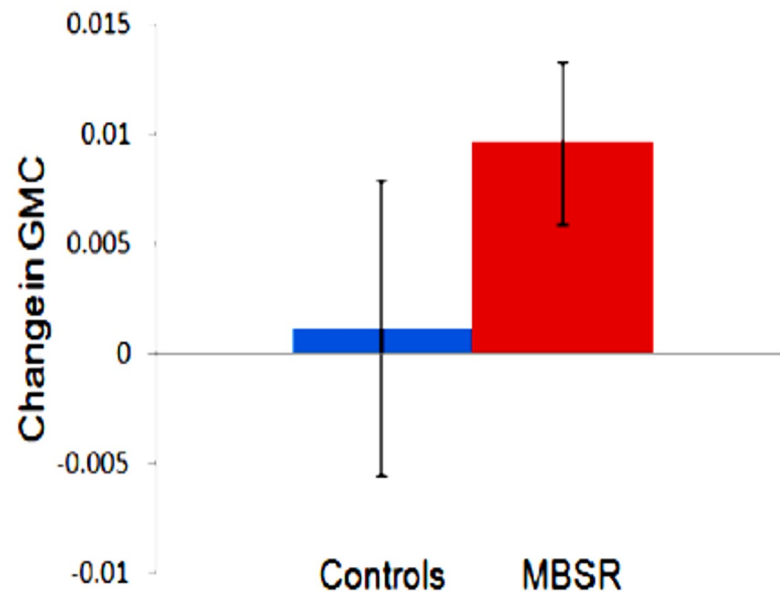


Lazar et al. 2005

Mindfulness and brain thickness



Left hippocampus



➤ Hippocampal thickening after 8 weeks of MBSR



ARE WE REALLY FREE? NOVEL INSIGHTS FROM BEHAVIORAL NEUROSCIENCE

Pietro Pietrini

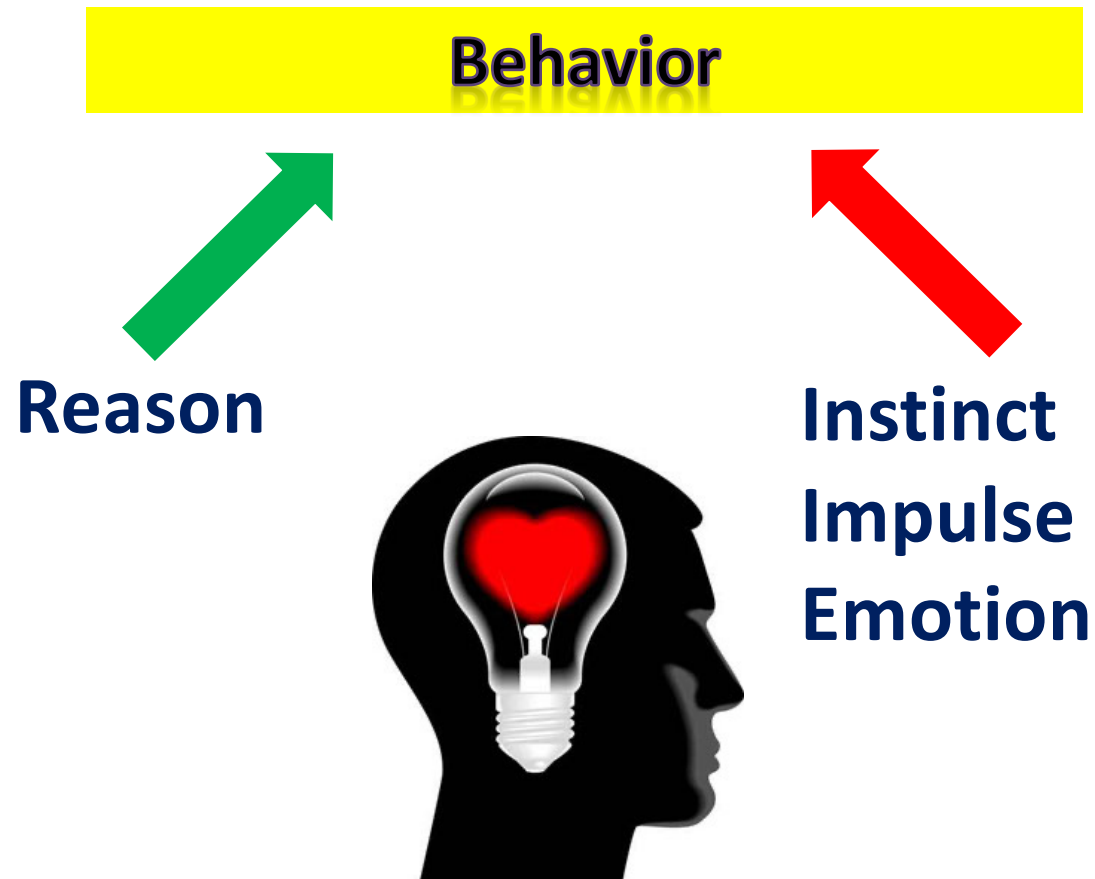
Chief, *Molecular MInd Lab*
IMT School for Advanced Studies Lucca

pietro.pietrini@imtlucca.it

What drives our behavior?

How do we make a decision?

How do we make a decision to act?



What Drives Behavior in Animals?





**Bears will eat just
the skin of the
wild salmon**



**Bears will eat just
the skin of the
wild salmon...**

**... and throw away
the delicious and
healthy Salmon
meat**

WHY?



Bears will eat just the **fat** skin of the wild salmon...

... and throw away the delicious, **lean** and healthy Salmon meat

WHY?



**Genetically
programmed to
hibernate → need
Brown Fat to
sustain metabolic
needs for months**

**So Bears are
genetically
programmed to
prefer fat over
lean food**



Which factors modulate *animal* behavior?

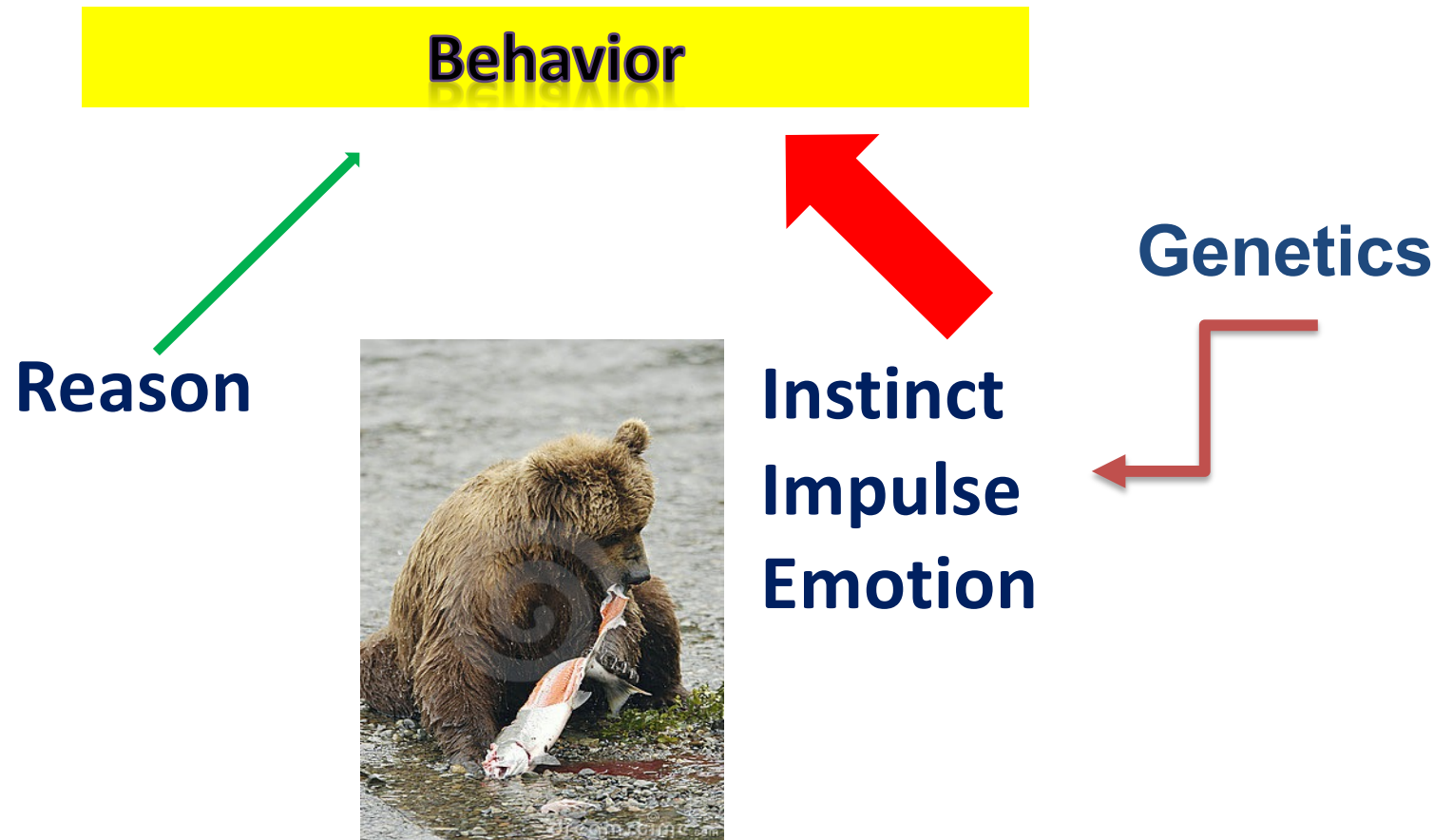
Behavior



**Instinct
Impulse
Emotion**

Genetics

Which factors modulate **animal** behavior?



Animals can control their behavior as well



**Are We Humans *THAT* Different from
Bears?**

Are We Humans *THAT* Different from Bears?

Which one would you pick?

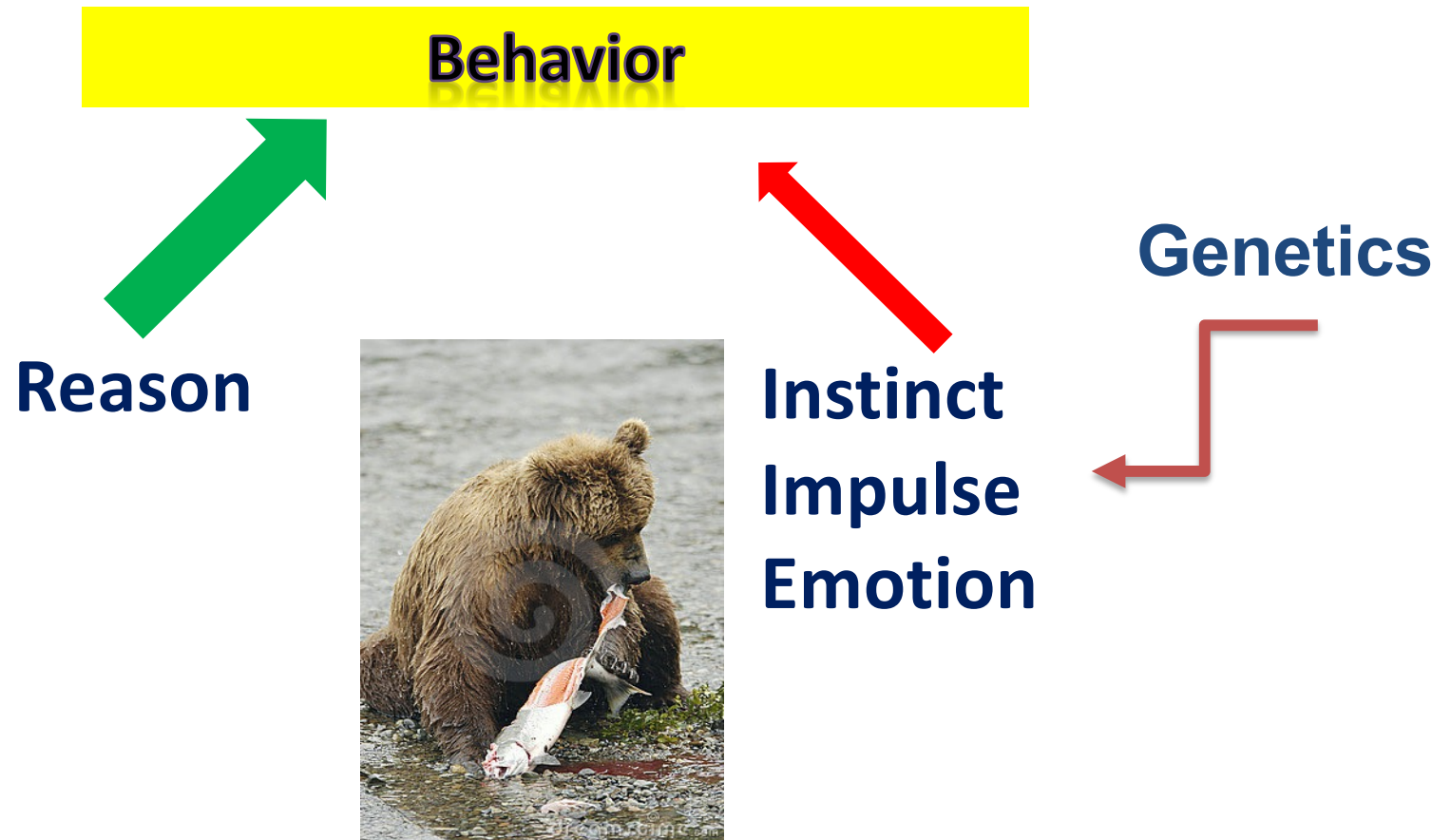


Are We Humans *THAT* Different from Bears?

Which one would you pick?

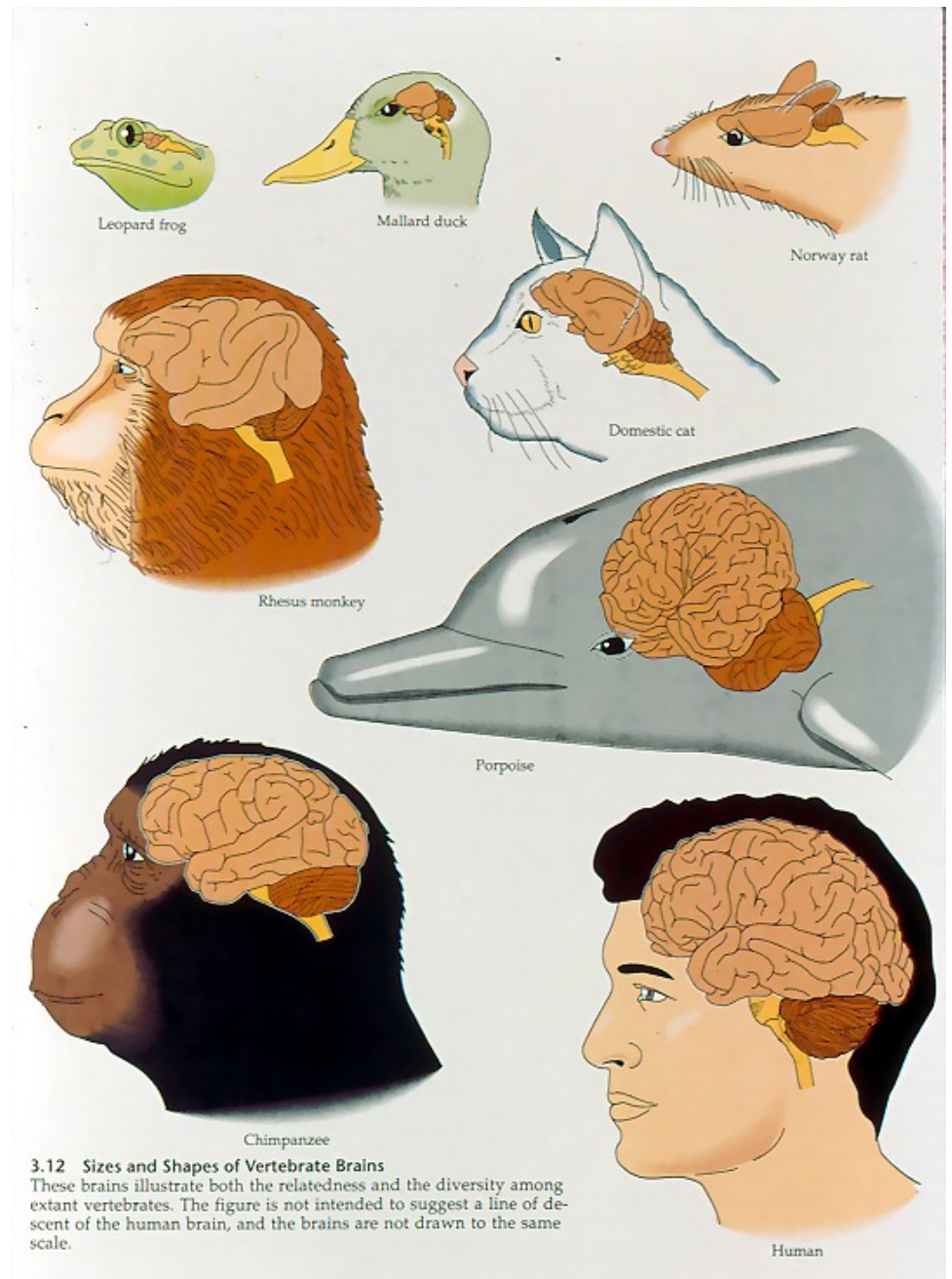


Which factors modulate *human* behavior?

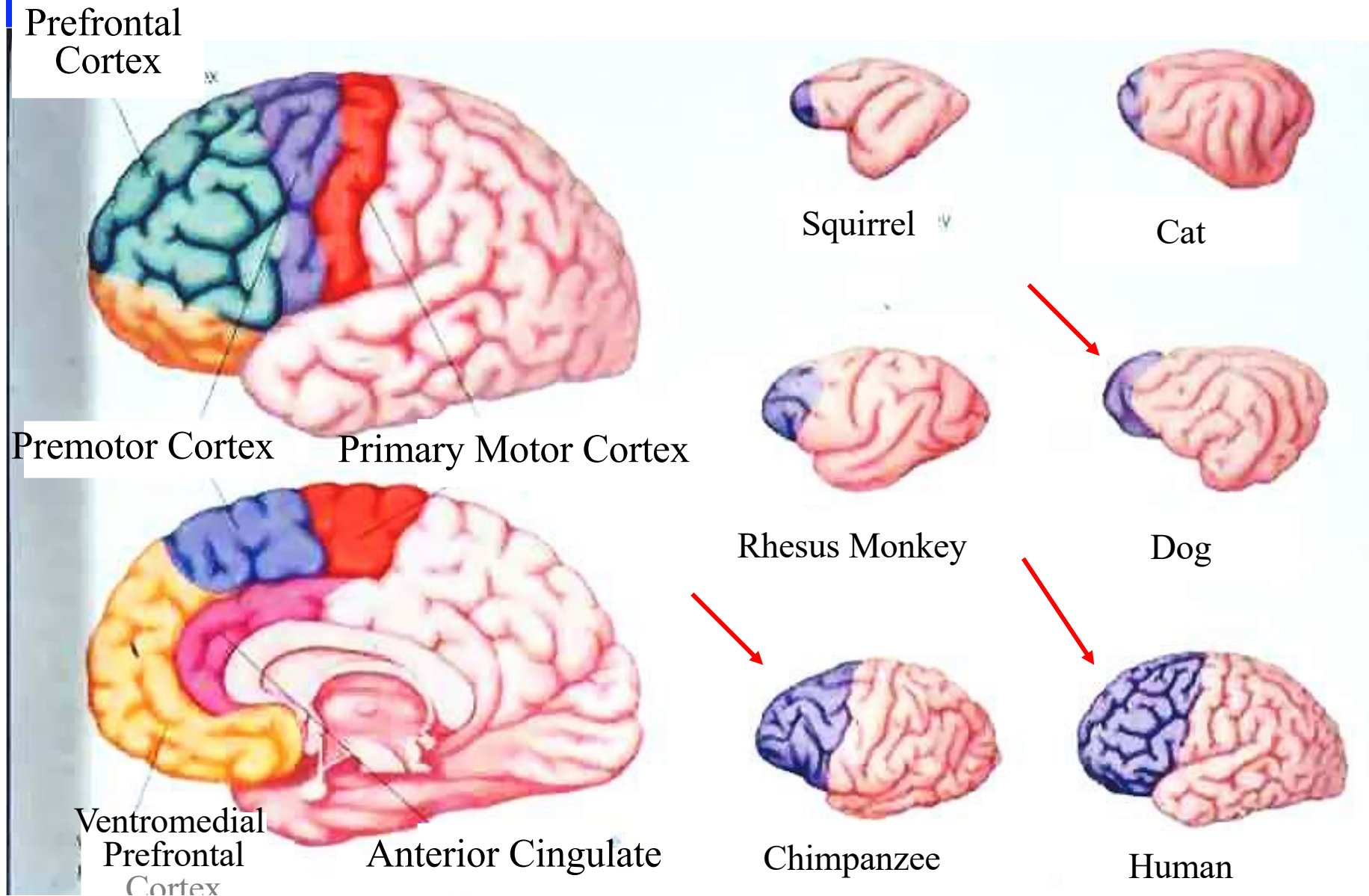


Exploring the Neurobiological Roots of Behavior

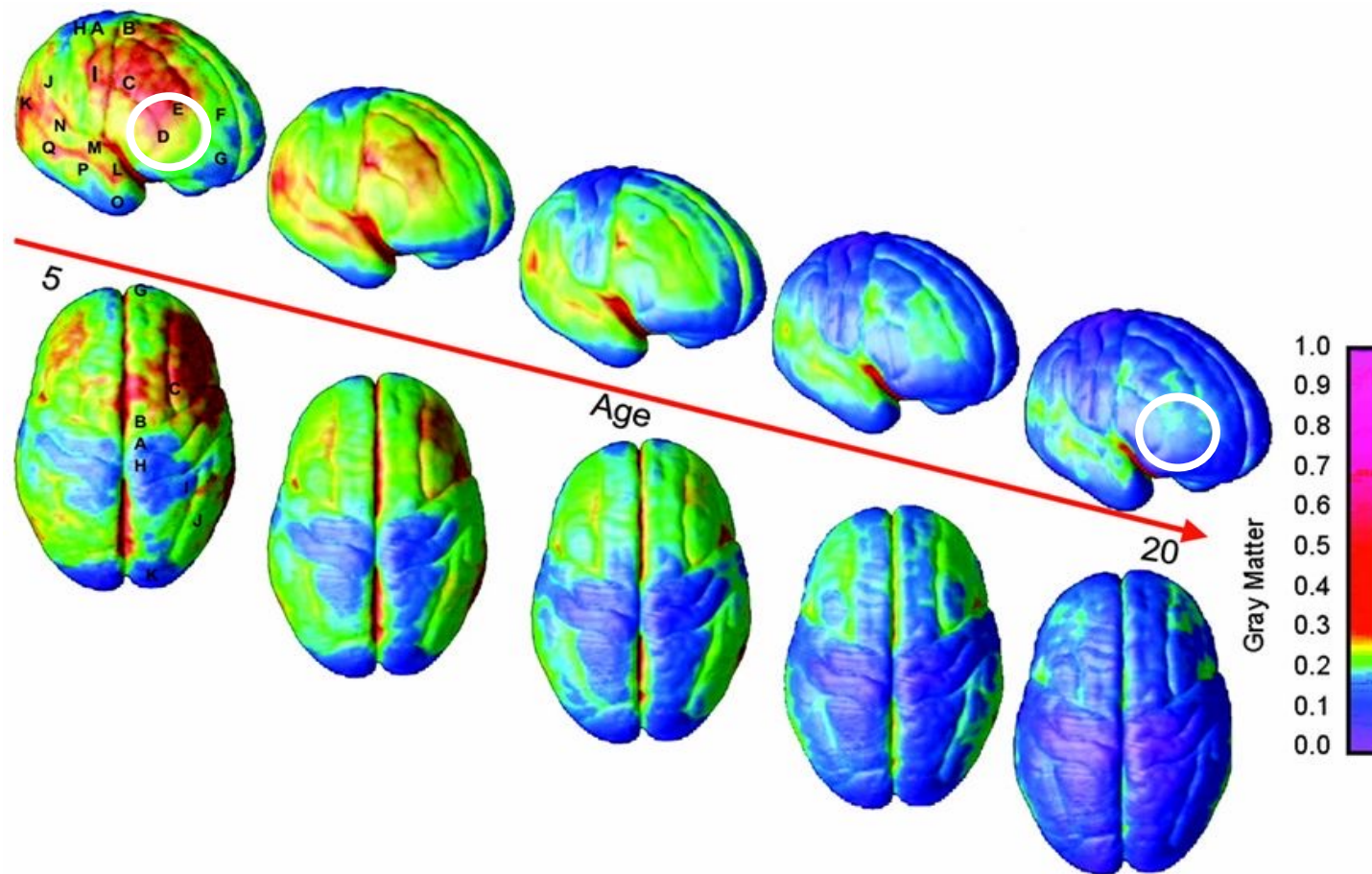
- **Structural and Functional Exploration of the Brain:**
a “true window” on the brain morphological and functional architecture
- **Molecular Biology and Behavioral Genetics:**
how our genes (DNA) shape who we become



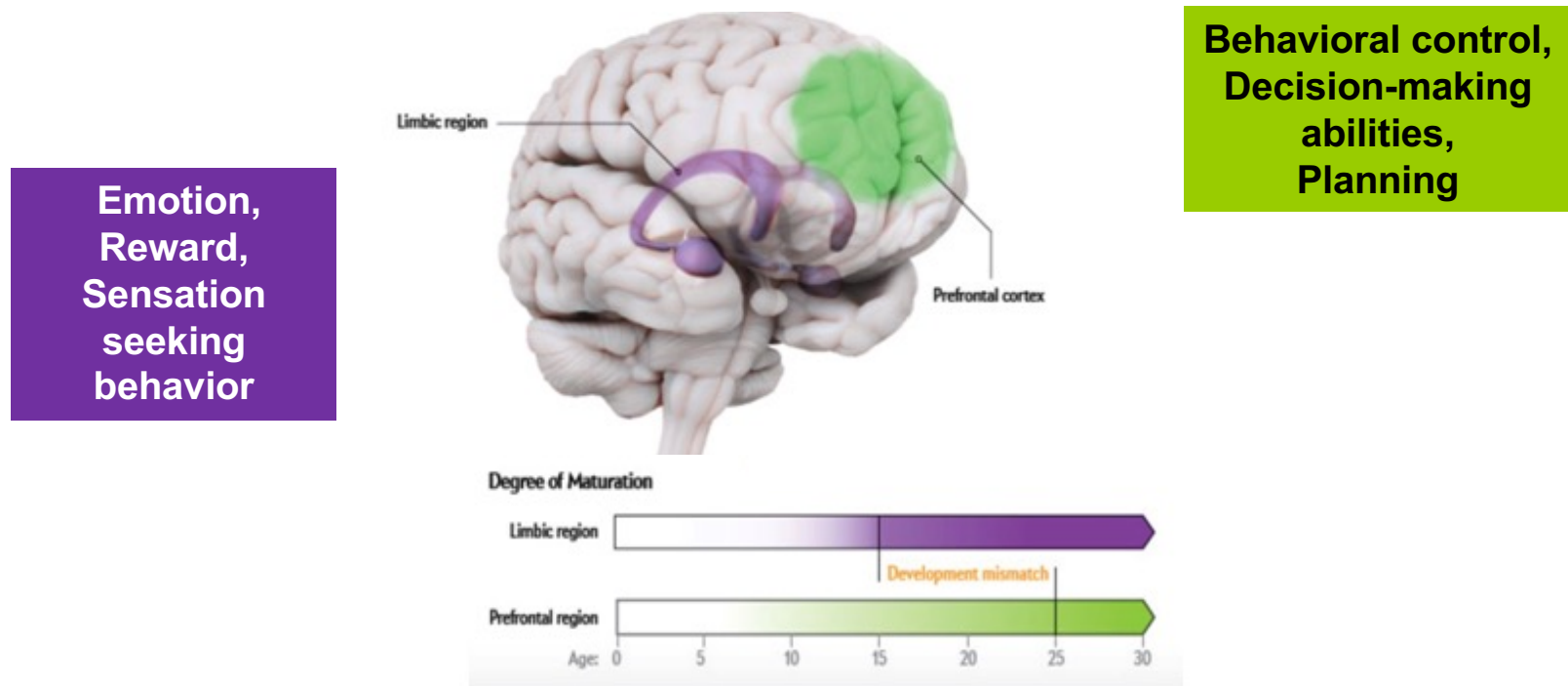
Brain Development and Functional Organization



Prefrontal cortex is the last to mature and reaches full maturation well after adolescence



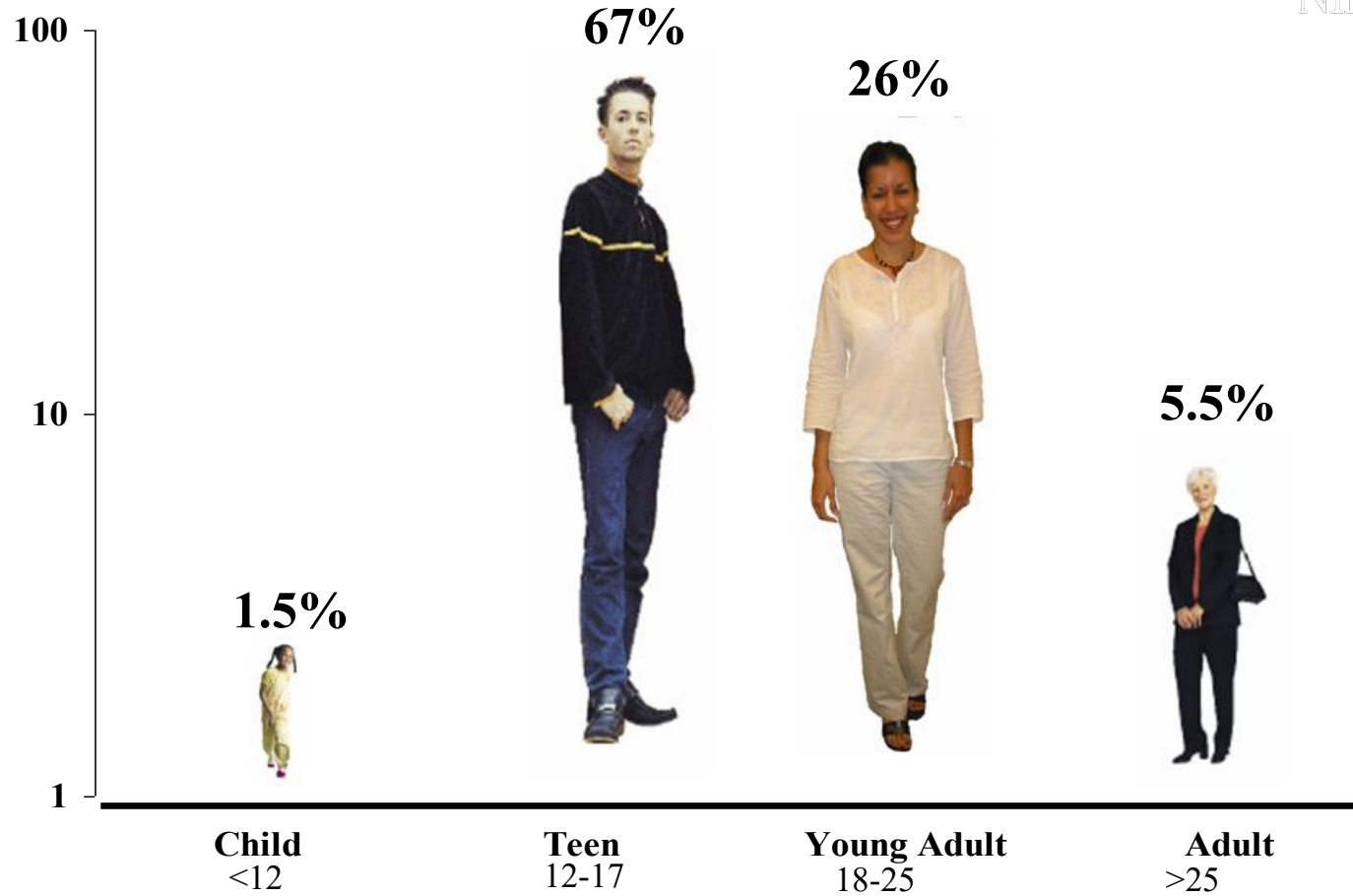
In Adolescence instinct tends to prevail over reasoning



Ten years temporal shift between maturation of the limbic system and the later maturation of the prefrontal cortex

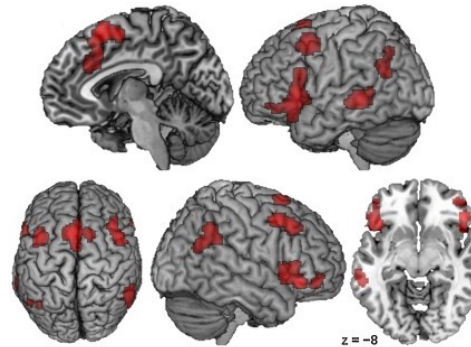
Dependence disorders begins in Adolescence

NIDA



How much are we really in
control of our behavior?

Suppressing spontaneous emotional responses

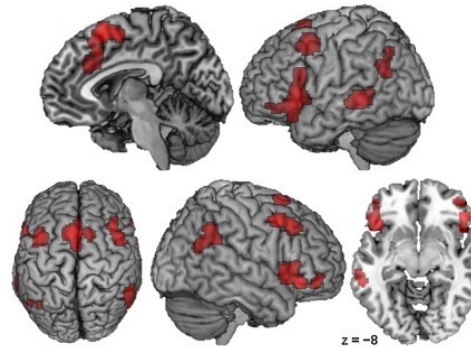


Langner et al., Neurosci & Biobehav Rev, 2018

Emotional responses are controlled by a parieto-temporal,
frontomedial and insula network

Avvenuti et al., J Cogn Neurosci, 2021

What happens when we feel like laughing but we are not allowed to?

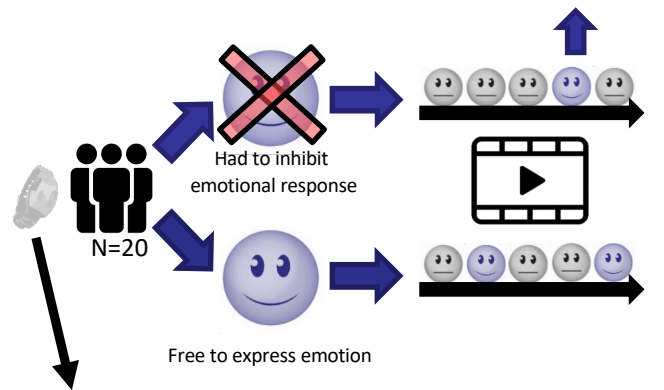


*Langner et al., Neurosci
& Biobehav Rev, 2018*

Avvenuti et al., J Cogn Neurosci, 2021

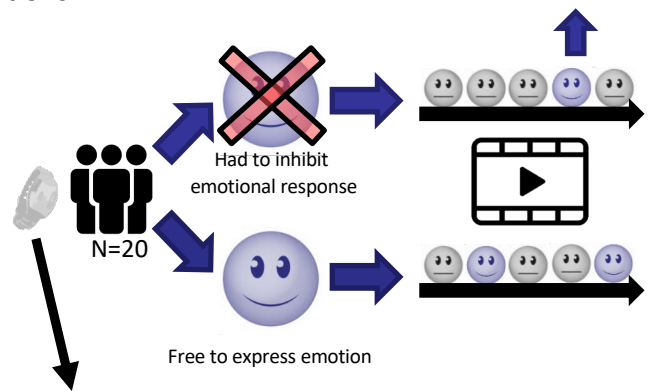
Do Not laugh while watching a funny movie!

- 15 minute funny movie
- Half of the individuals in the group had to keep a **neutral facial expression**
- The other half could express their emotions



Do Not laugh while watching a funny movie!

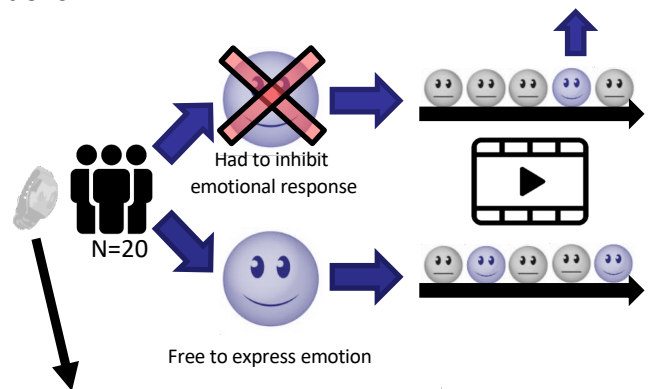
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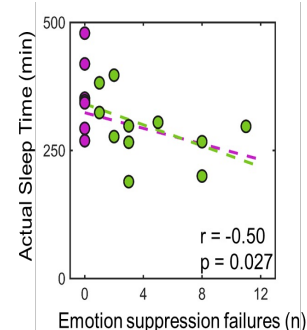
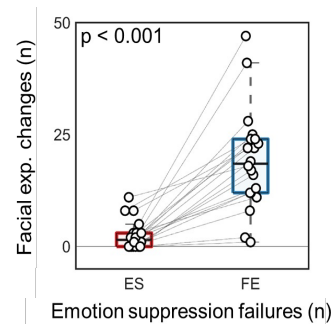
Sleep duration the night before the experiment was measured by **actigraphy**

(Lack of) sleep and emotional response control

- 15 minute funny movie
- Half of the individuals in the group had to keep a **neutral facial expression**
- The other half could express their emotions



Sleep duration the night before the experiment was measured by actigraphy

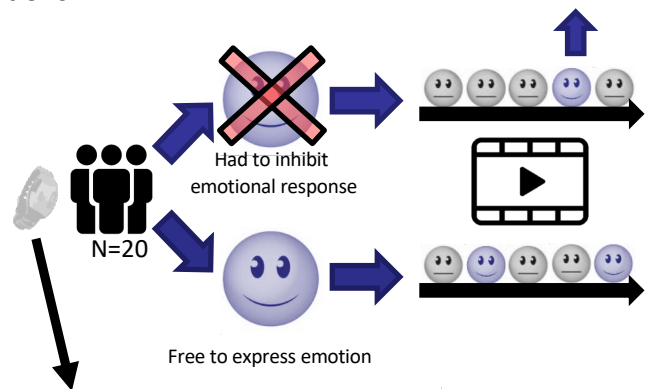


Emotion Suppression Failures
were inversely correlated to
sleep time

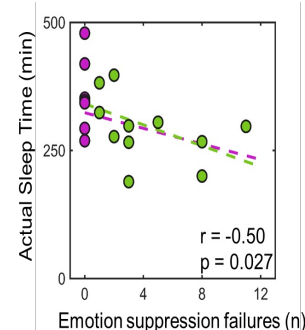
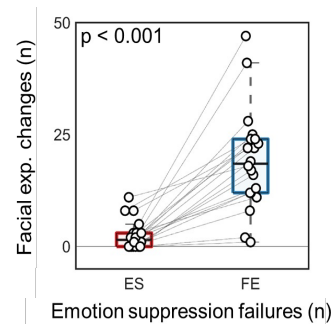


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Ability to control emotions is impaired when we are tired

Emotion Suppression Failures
were inversely correlated to
sleep time



Does Fatigue
Affect the
Decision Making Process?

Extraneous factors in judicial decisions

Shai Danziger, Jonathan Levav , and Liora Avnaim-Pesso [Authors Info & Affiliations](#)

Edited* by Daniel Kahneman, Princeton University, Princeton, NJ, and approved February 25, 2011 (received for review December 8, 2010)

April 11, 2011 | 108 (17) 6889-6892 | <https://doi.org/10.1073/pnas.1018033108>

- Is “*what the judge ate for breakfast*” relevant in the parole decision making process?
- Authors recorded the judges’ two daily food breaks, segmenting the deliberations of the day into **three distinct “decision sessions”**
- **1,112 judicial rulings**, collected over 50 day in a 10-mo period, by **eight Jewish-Israeli judges** (two females)
- Prisoner sample
 - 727 Jewish-Israeli males (65.3%)
 - 326 Arab- Israeli males (29.3%)
 - 50 Jewish-Israeli females (4.5%)
 - 9 Arab-Israeli females (0.9%).
- Crimes were **embezzlement, assault, theft, murder and rape**
- Each parole board was composed of one judge, a criminologist and a social worker

Extraneous factors in judicial decisions

Shai Danziger, Jonathan Levav , and Liora Avnaim-Pesso [Authors Info & Affiliations](#)

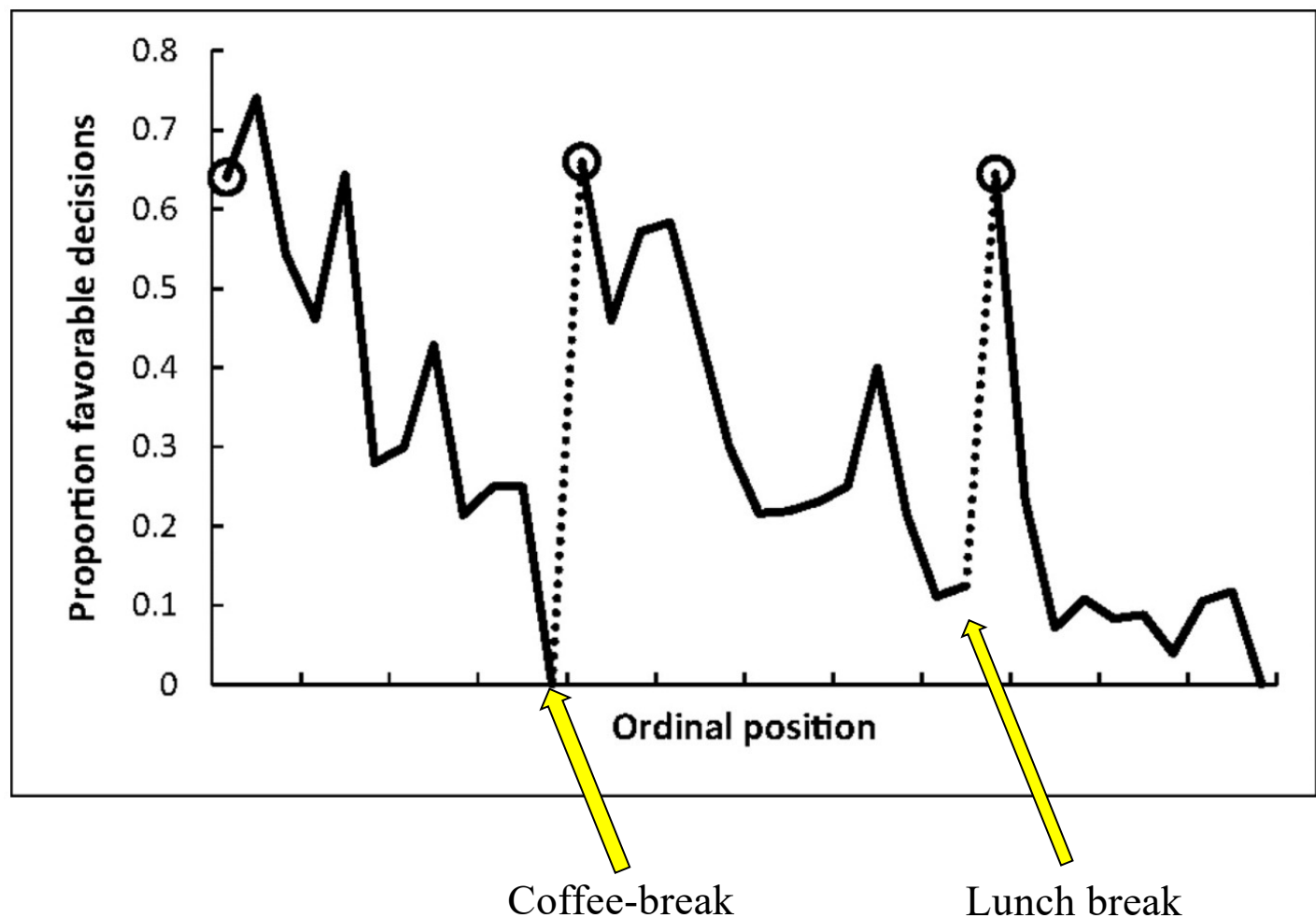
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Favorable rulings dropped gradually from $\approx 65\%$ to nearly zero within each decision session

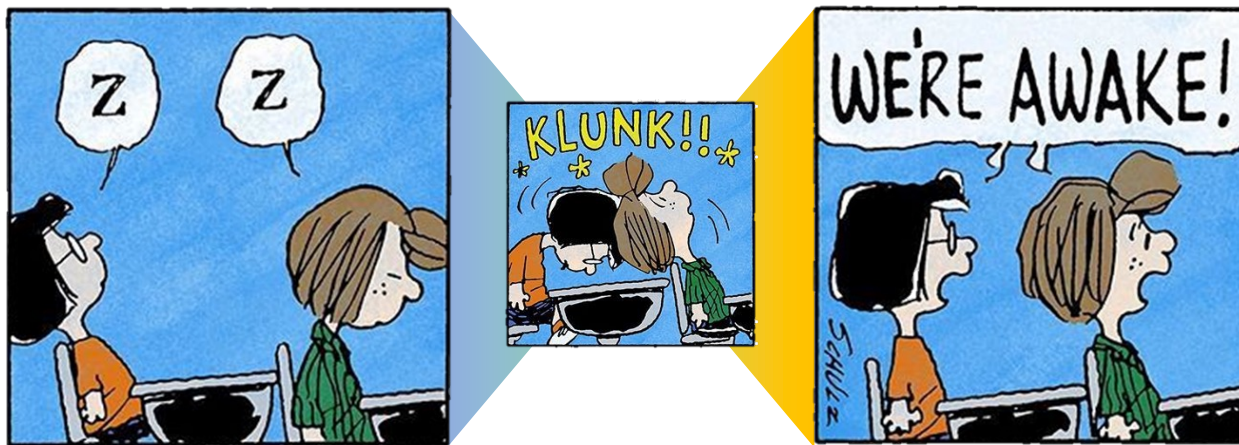
and

returned abruptly to $\approx 65\%$ after each break.



What happens in the brain when
we are fatigued?

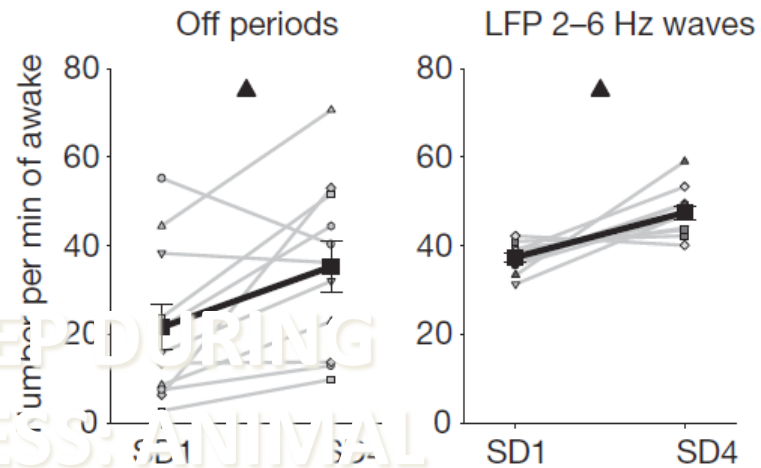
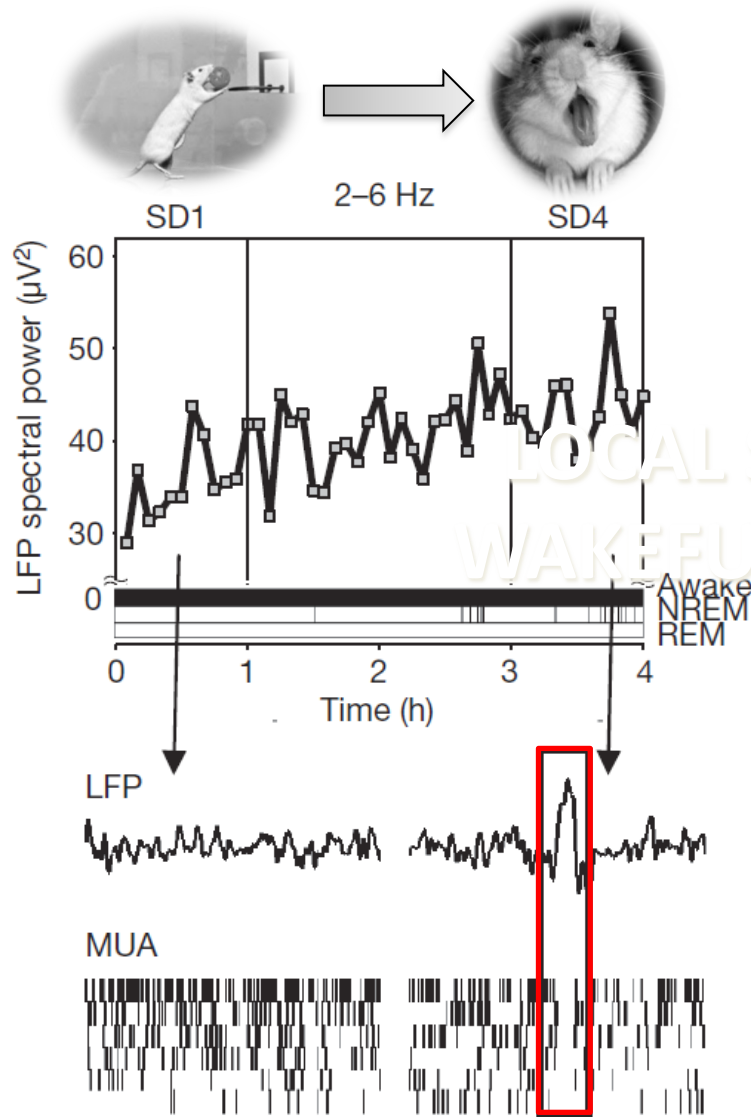
Awake, asleep... or both



Wakefulness and sleep have traditionally been regarded as two mutually exclusive states

But is this really the case?

Local Sleep in the Awake Rat



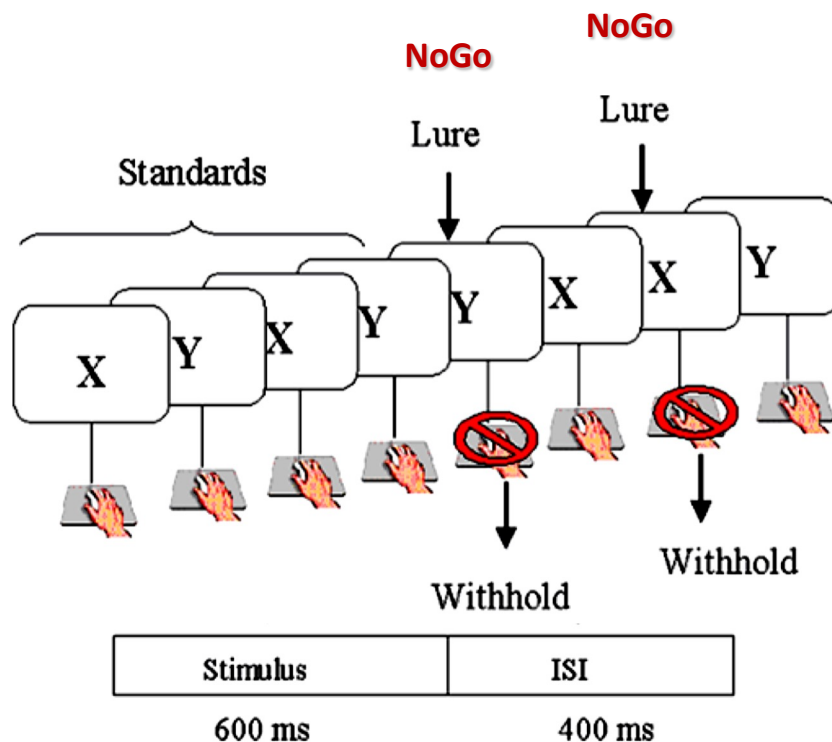
After a long period in an awake state, cortical neurons can go briefly 'offline' as in sleep, accompanied by **slow waves** in the local EEG

Neurons often go offline in **one cortical area** but not in another [**Local Sleep**]

Local sleep and impulse control



Correlation between potential local sleep episodes (i.e., a temporary slowing down of EEG activity) and behavioral performance during an impulse control task



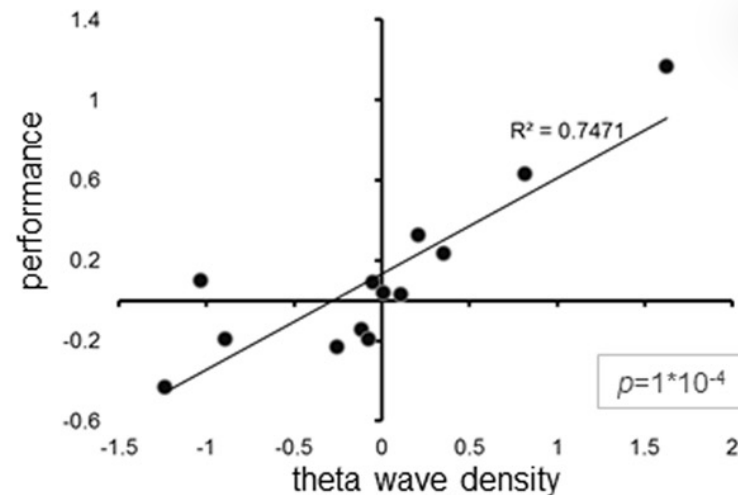
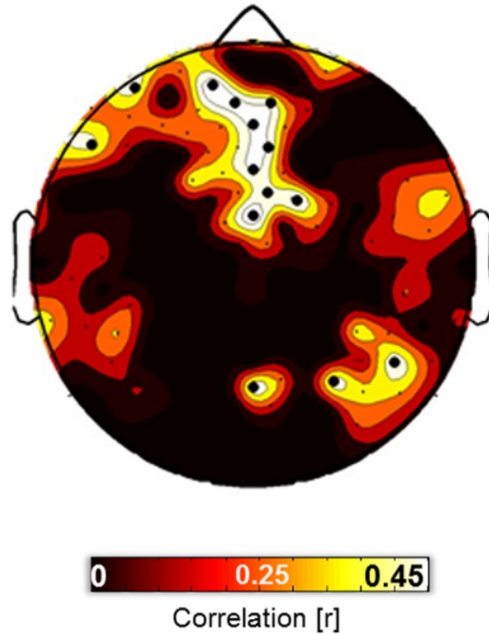
	Response	No Response
GO	Correct Response (CR)	Lapse (RT>500ms) (LA)
NoGO	Commission Error (CE)	Correct Withhold (CW)

Local Sleep Like Activity and Behavior

Response Inhibition Test

– Commission Errors –

Pressing the button when supposed to withhold



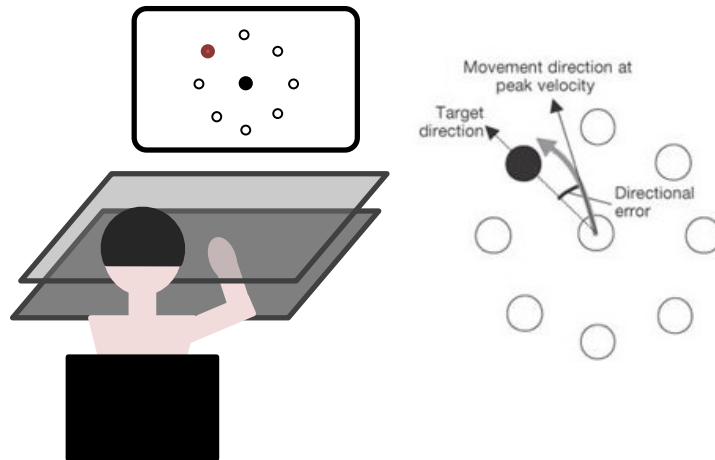
A higher number of EEG low-frequency THETA waves was associated with a relative impairment in impulse control

Local Sleep Like Activity and Behavior

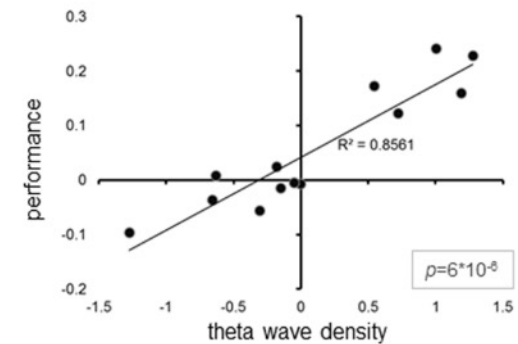
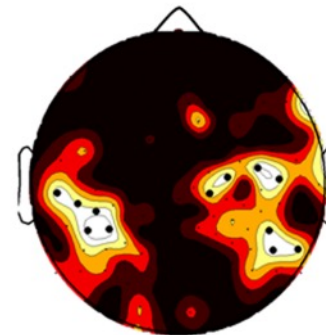
Motor Reaching Test



Motor Reaching Test



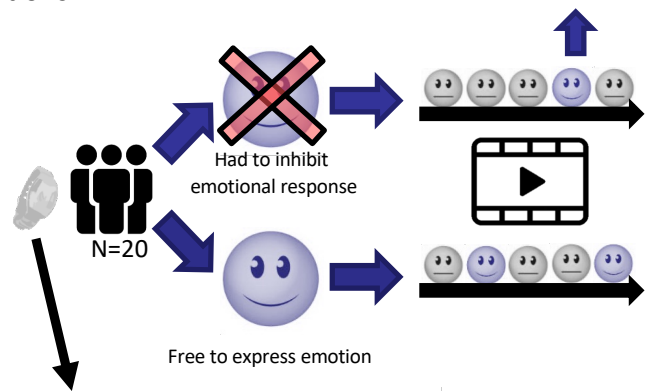
Movement Time



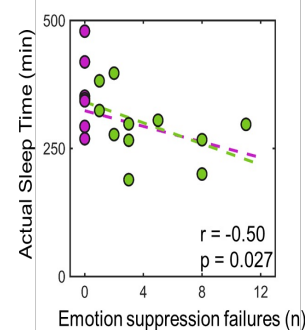
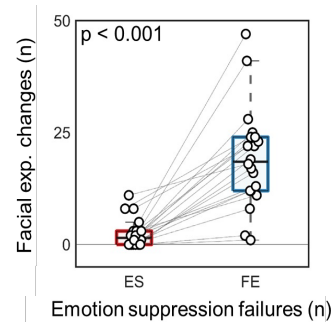
A higher number of EEG low-frequency THETA waves was associated with a relative impairment in motor reaching

(Lack of) of sleep and emotional response control

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- Half of the individuals in the group had to keep a **neutral facial expression**
- The other half could express their emotions



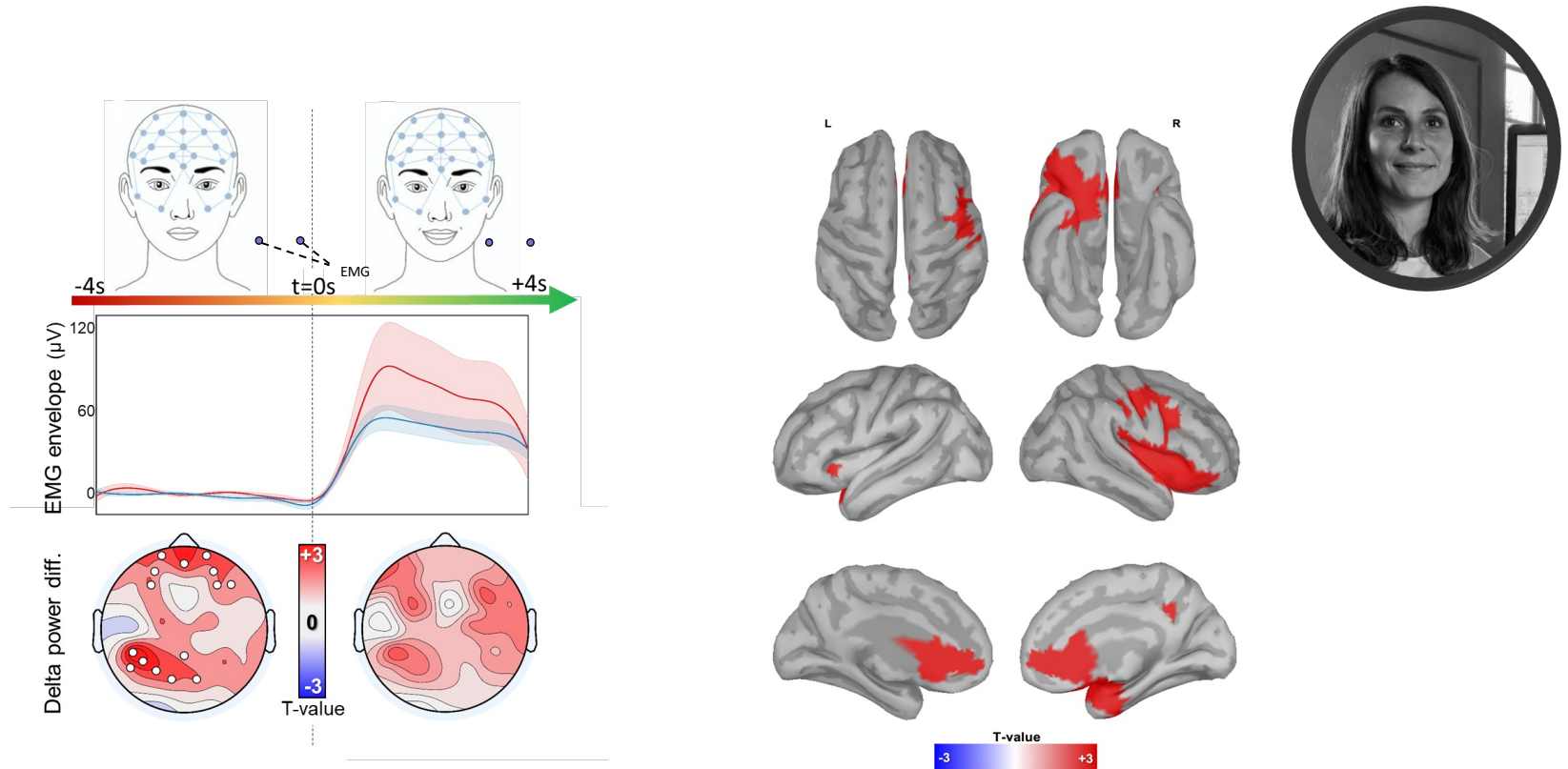
Sleep duration the night before the experiment was measured by actigraphy



Ability to control emotions is impaired when we are tired

Emotion Suppression Failures were inversely correlated to sleep time the night before experiment

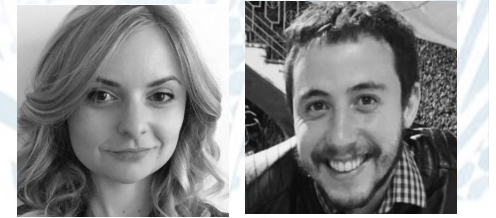
Brain areas that control our emotional response “fall asleep”



Failure to control emotional responses was associated to onset of EEG sleep-like activity in the parieto-temporal frontomedial insula network

Avvenuti et al., J Cogn Neurosci, 2021

Does Fatigue
Affect the
Social Behavior?



Prolonged exertion of self-control causes increased sleep-like frontal brain activity and changes in aggressivity and punishment

Erica Ordali^{a,1,2} , Pablo Marcos-Prieto^{b,1,2}, Giulia Avenuti^a , Emiliano Ricciardi^a , Leonardo Boncinelli^c , Pietro Pietrini^a , Giulio Bernardi^{a,1} , and Ennio Bilancini^{b,1} 

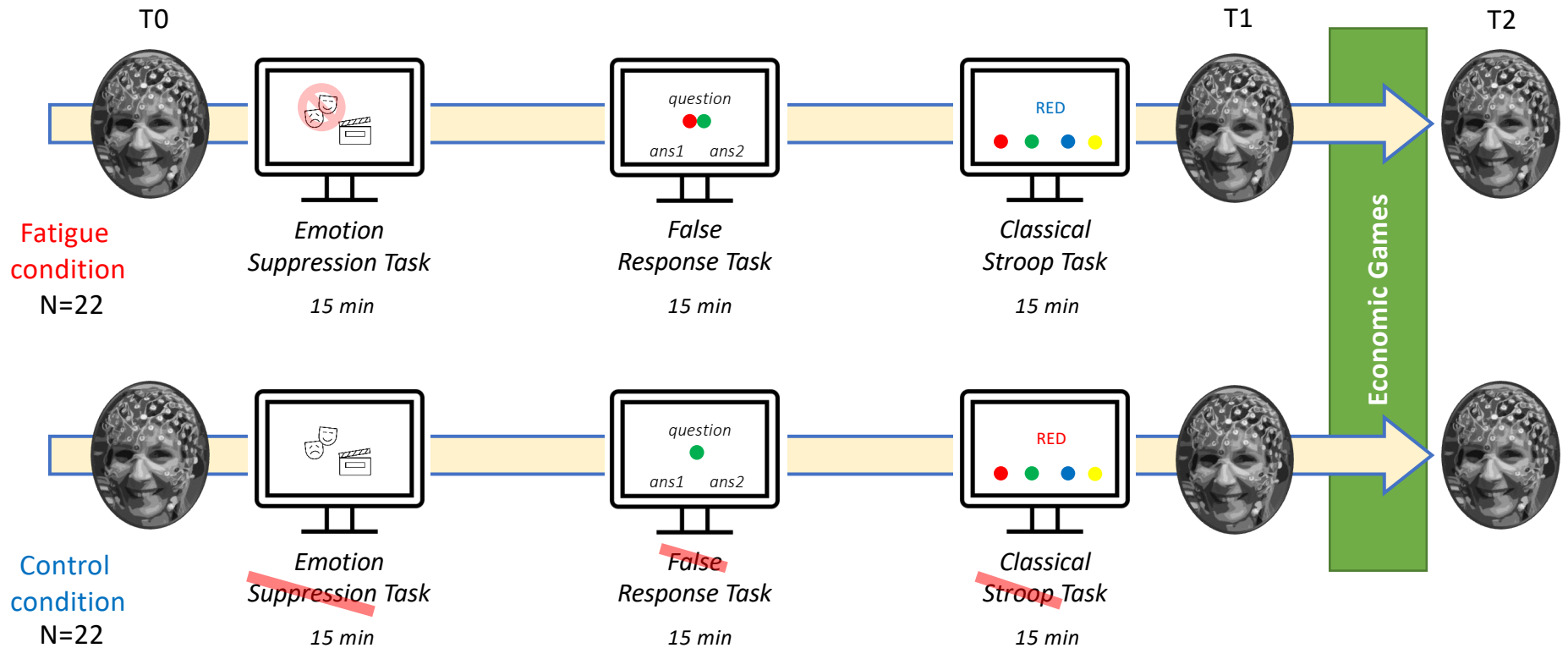
Affiliations are included on p. 5.

Edited by Roy F. Baumeister, The University of Queensland, Brisbane, QLD, Australia; received March 11, 2024; accepted October 8, 2024, by Editorial Board Member Michael S. Gazzaniga

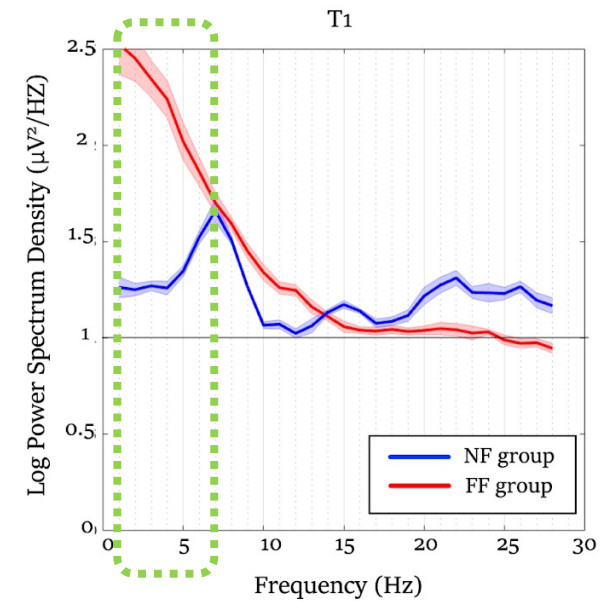
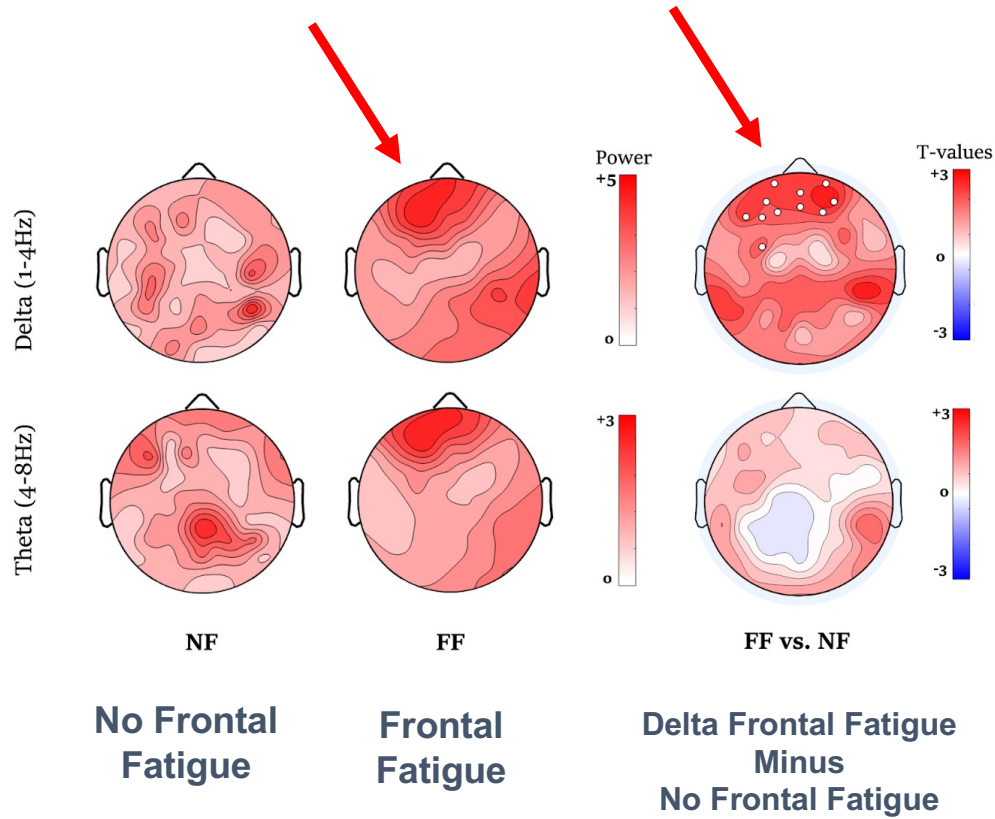
Significance

In this paper, we demonstrate that prolonged exertion of self-control via cognitively demanding tasks induces a state of fatigue marked by the emergence of sleep-like brain activity within the prefrontal cortex. While they were in this state, individuals displayed an increased propensity to behave aggressively during economic games that simulated socially relevant scenarios. Specifically, we observed a heightened hostility in the Hawk and Dove game and a marked tendency for spiteful punishment in the Public Goods Game. These findings indicate that the propensity for prosocial behavior may be reduced in states of cognitive fatigue resulting from the extended exertion of self-control

Local sleep-like activity and social behavior in humans



Local sleep-like activity and social behavior in humans



Local sleep-like activity and social behavior in humans

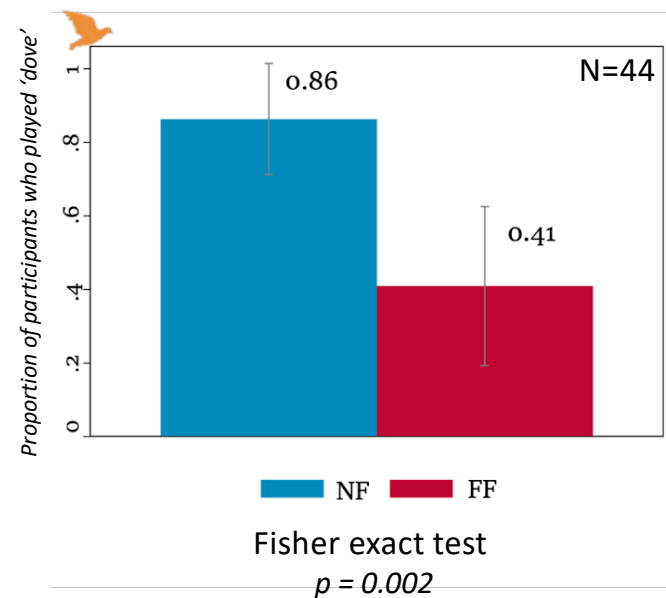
Hawk-Dove Model: Costs and Benefits of Fighting over Resources

Payoff* to...	...in fights against:	
	hawk	dove
hawk	Hawk wins 50% of fights; is injured in 50% of fights. Payoff: $(V-D)/2$	Hawk always wins; dove flees. Payoff: V
dove	Dove never wins; is never injured. Payoff: 0	Dove wins 50% of fights; is never injured; wastes time. Payoff: $V/2 - T$

*V = fitness value of winning resources in fight

D = fitness costs of injury

T = fitness costs of wasting time



Local sleep-like activity and social behavior in humans

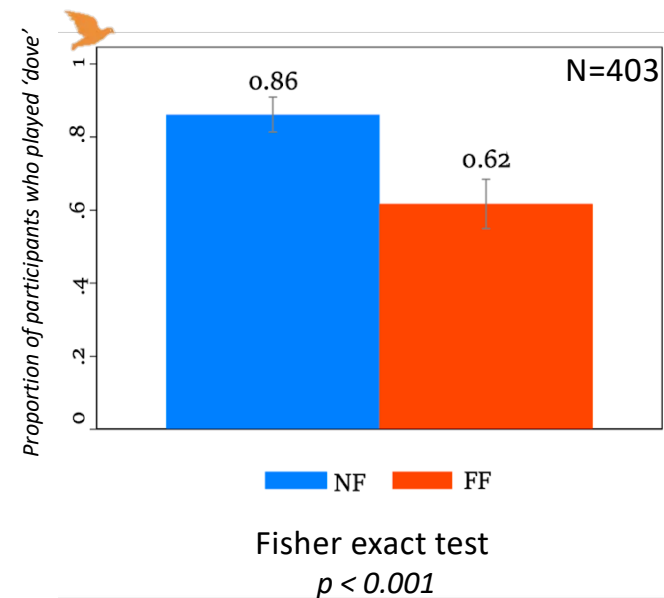
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Implications for everyday social life and work

- Brain Fatigue appears after a relative short time of cognitive engagement



Implications for everyday social life and work

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- Brain Fatigue affects task-related areas and impairs performance



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Implications for everyday social life and work

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Implications for everyday social life and work

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- Overall, these findings carry implications for scheduling work meetings, school teaching or judicial mediations



Implications for everyday social life and work

- Brain Fatigue appears after a relative short time of cognitive engagement  
- Brain Fatigue affects task-related areas and impairs performance
- Prolonged self-control leads to Frontal Fatigue
- Frontal fatigue favors aggressive behavior in social interactions
- Overall, these findings carry implications for scheduling work meetings, school teaching or judicial mediations
- Science appears to provide support to popular wisdom says like “
 - *sleep on it – dormici sopra*
 - *the night brings counsel – la notte porta consiglio*
 - *The early bird catches the worm - la mattina ha l'oro in bocca*

Implications for Forensic Psychiatry and the Law

**How do we control impulsive/aggressive
behavior?**

MENTAL CAPACITY FOR THE LAW («SOUND MIND»)

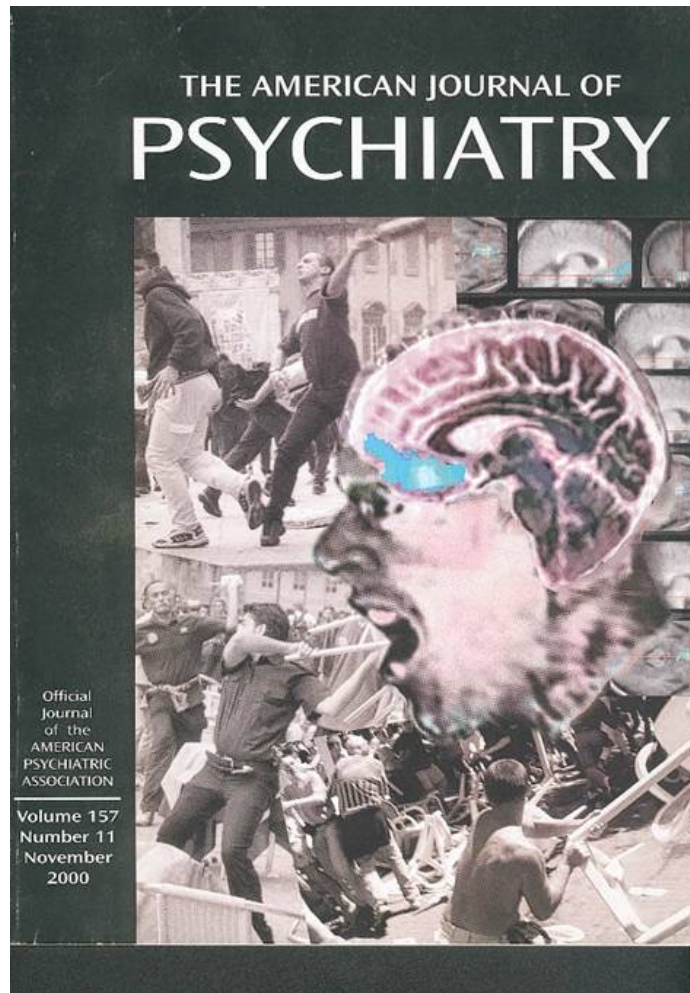
- **Capacity to understand (Capacità di intendere)**

The ability to understand the nature of one own's actions and their consequences

- **Capacity to will (Capacità di volere)**

The ability to control one own's behavior.

Functional neuroanatomy of behavior – Aggression



**ANATOMO-FUNCTIONAL
CORRELATES WITHIN
FRONTAL LOBES**



- Impulse and aggression discontrol
- Impaired decision-making process
- Impaired critical ability and judgement

Article

Neural Correlates of Imaginal Aggressive Behavior Assessed by Positron Emission Tomography in Healthy Subjects

Pietro Pietrini, M.D., Ph.D.
Mario Guazzelli, M.D.
Gianpaolo Basso, M.D.
Karen Jaffe, B.A.
Jordan Grafman, Ph.D.

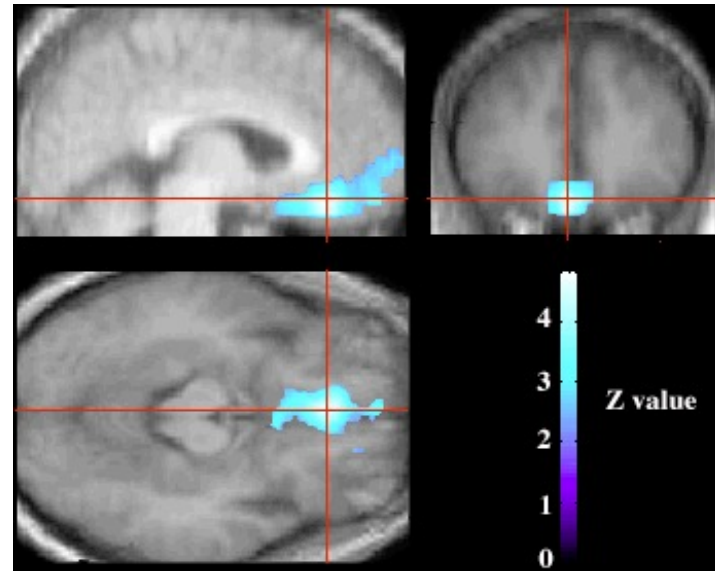
Objective: Neurodegenerative or traumatic lesions of the frontal lobes often lead to abnormally aggressive behavior. The authors hypothesized that the imaginal evoking of scenarios involving aggressive behavior would be associated with a modulation of the functional activity in the human frontal cortex.

Method: Regional cerebral blood flow (rCBF) determinations by positron emission tomography and psychophysiological measures of emotional responsiveness were obtained in a group of 15 young healthy volunteers with good visual imagery abilities and no history of abnormal behavior while they imagined the same scenario with four variations involving emotionally neutral behavior and aggressive behavior.

Results: Compared to the imagined neutral scenario, the imagined scenario involving aggressive behavior were associated with significant emotional reactivity and rCBF reductions in the ventromedial prefrontal cortex, suggesting that a functional deactivation of this cortical area occurs when individuals respond to the eliciting of imagined aggressive behavior.

Conclusions: These results in healthy subjects further expand previous findings from animal and human studies by providing an in vivo functional demonstration of the involvement of the orbitofrontal cortex in the expression of aggressive behavior. They are also consistent with the hypothesis that a functional alteration of this cortical region may be present in individuals with pathological aggressive behavior.

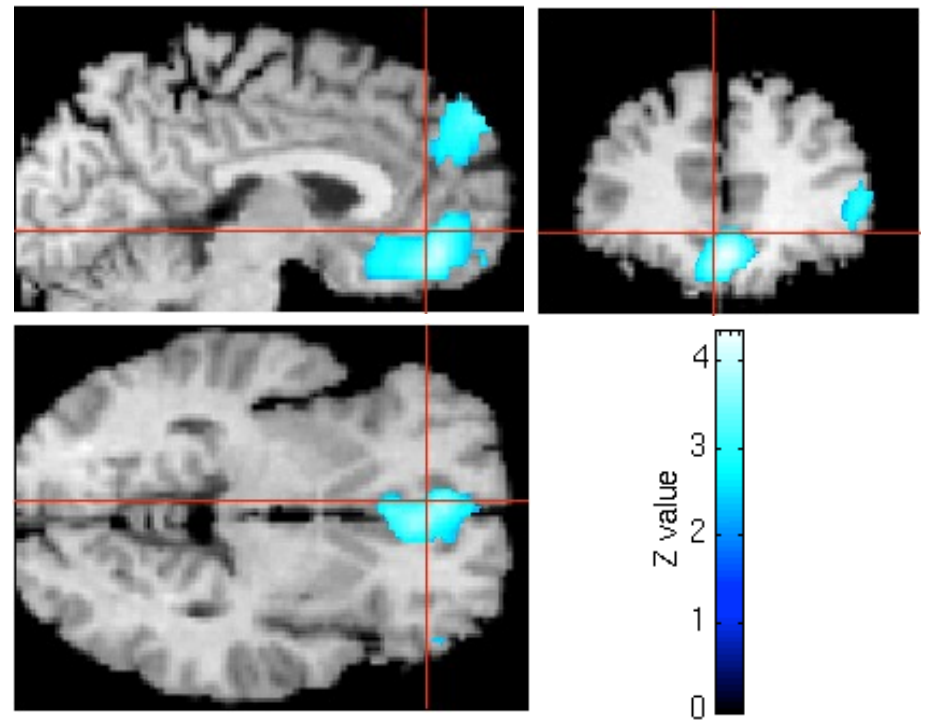
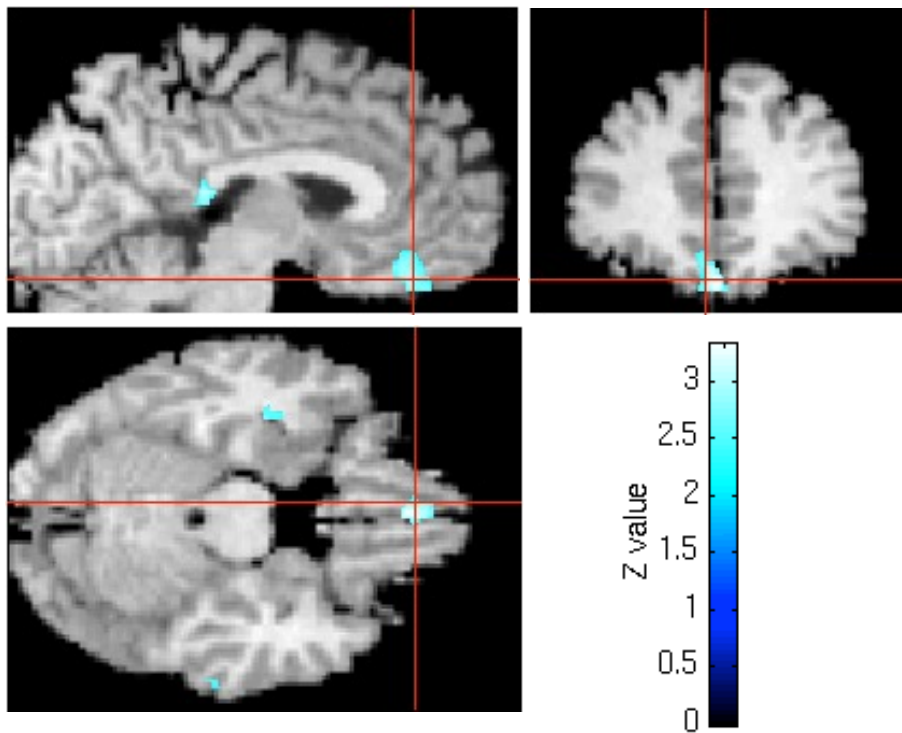
(Am J Psychiatry 2000; 157:1772-1781)



Functional suppression of prefrontal cortex during evocation of aggressive behavior is greater in females than males

Males

Females



$p < 0.01$

Males Have a Greater Attitude towards Violence than Females

	Whole Group	Males	Females
N	15	8	7
Age, yr	22 ± 2	22 ± 3	22 ± 1
Education	15 ± 1	16 ± 1	15 ± 1
Betts score	71 ± 1	74 ± 22	69 ± 17
Gordon score	23 ± 2	23 ± 2	23 ± 1
ATV score	67 ± 23	84 ± 22	51 ± 6 *
BDHI	32 ± 10	32 ± 11	32 ± 10
Self-esteem	25 ± 4	23 ± 4	26 ± 3
Body-esteem	45 ± 8	47 ± 6	43 ± 10
Anger Inventory	191 ± 34	186 ± 27	196 ± 40

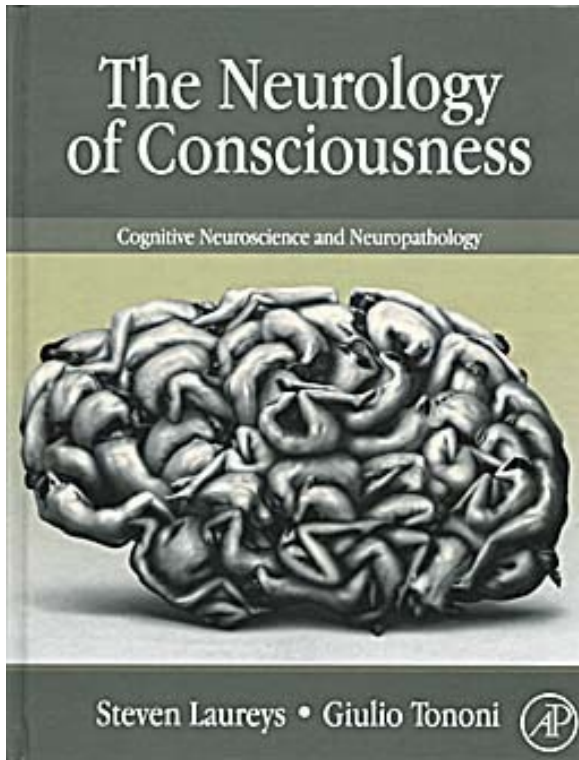
Betts Questionnaire of Mental Imagery Vividness Scale: score range 35 (best) to 235 (poor)

Gordon Test of Visual Imagery Control: max best score 24

ATV: Attitude Towards Violence (max score: 208); BDHI: Buss-Durkee Hostility Inventory (max score: 75)

Self-esteem scale: maximum score (highest s. e.) = 30; Body-esteem scale: max score (highest b. e.) = 70

Anger Inventory: max score (highest anger) = 360

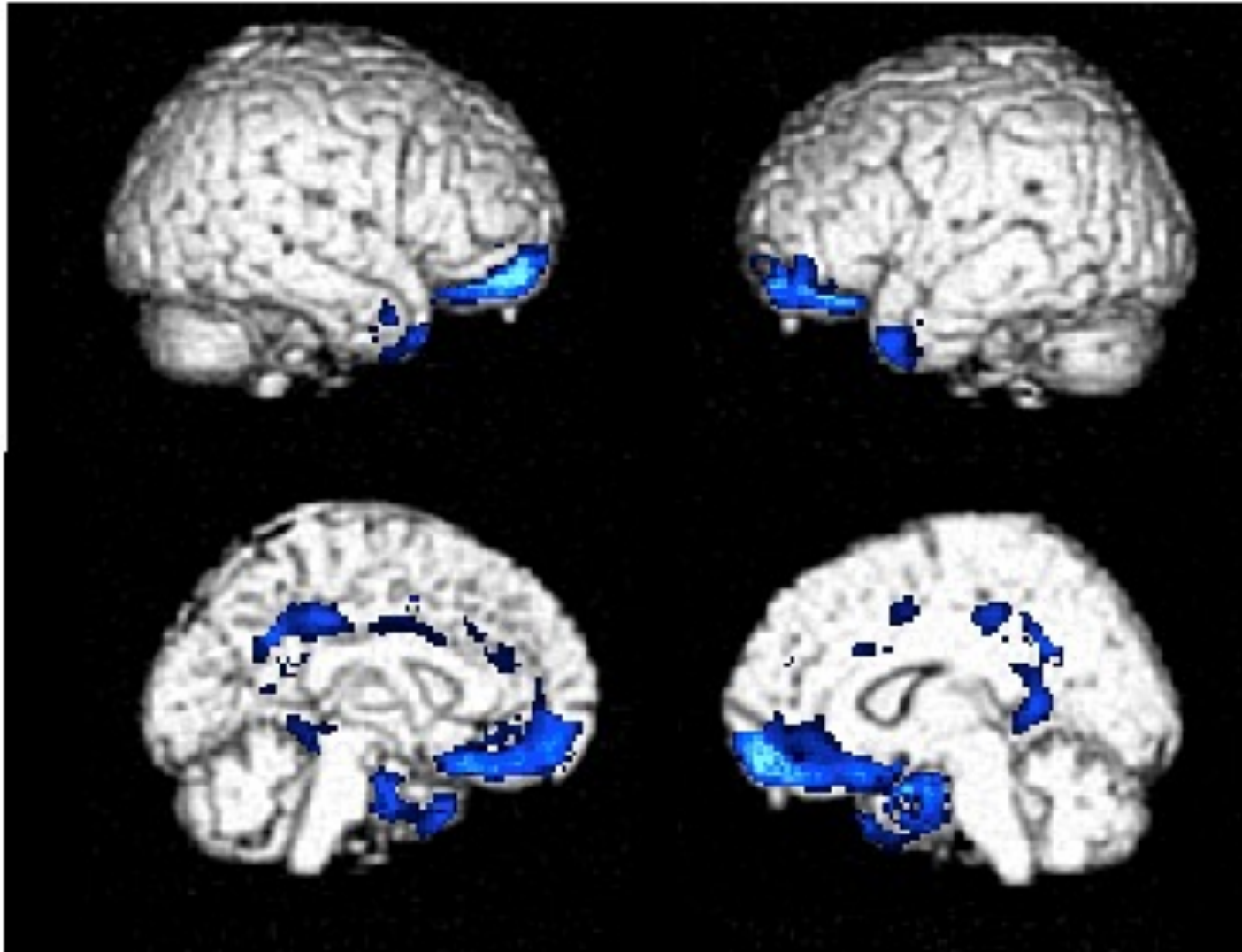


“You are your brain”

S. Laurey and G. Tononi,
The Neurology of Consciousness, p. IX, Elsevier, 2009

From Instinct to Behavior.....

Brain gray matter abnormalities in adult males with psychopathy (n=257; Ermer et al., 2012)



Courtesy Prof. Kent Kiehl

Are Criminals BAD or MAD?



Antisocial Personality Disorder
Serial killers